

The middle and the ends I.

Uniform Bessel bounds for endpoint crossings in the persistent random walk

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Abstract

A ball falls through a Galton board and makes N steps, each to the left or to the right. The first step is fair, and each later step repeats the one before it with probability p . Kagey asked for which p the middle bin is exactly as likely as the 2 end bins. We answer this for every N . The steps form the zero-field Ising chain with free boundaries, and the answer is governed by $T = N(1 - p)$, essentially the expected number of direction changes. We compare the ratio of each bin to an end bin with an explicit Bessel function, with a relative error bound that holds at every N . At the center this traps the crossing value z_N of T in an explicit interval whose width tends to 0, for every $N \geq 2$. The proof is analytic for $N \geq 175$, uses exact integer certificates below that, and is also checked in Lean. As a consequence $z_N + \frac{1}{2} \log z_N = \log N + \frac{1}{2} \log(\pi/8) + 1/(8z_N) + R_N$ with an explicit bound on R_N . We also find the crossing of every other bin, compare free and periodic boundaries, and prove a local limit theorem for the bin probabilities whose relative error tends to 0 uniformly for $1 - p \leq \frac{1}{2}$, from the diffusive to the ballistic regime.

Keywords: persistent random walk; correlated random walk; Galton board; Markov binomial distribution; Ising chain; modified Bessel functions; local limit theorem; computer-assisted proof.
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1 Introduction

1.1 Kagey’s question

Suppose you drop a ball into a Galton board. At each peg it bounces left or right. On an ordinary board the bounces are fair and independent, the bin is binomial, and the 2 end bins are the least likely. Problem 131 of Kagey’s Open Problem Collection [1] gives the ball some momentum. The first bounce is fair, and after that each bounce goes the same way as the one before it with probability p . At $p = \frac{1}{2}$ this is the ordinary board. As p grows the ball tends to keep going, and the end bins gain weight.

Kagey asks for the law of the bins, for the meaning of the tables of numerators that this law produces for $p = \frac{2}{3}$ and $p = \frac{1}{3}$, for the value of p at which the middle bin and the end bins are equally likely, whether there is a normal limit, and what changes on a cylinder. The third question is the main one, and most of this paper is about it.

Here are the smallest cases. With 2 steps there are 3 bins. The ball lands in the middle exactly when it turns once, so the bins have probabilities $\frac{p}{2}$, $1 - p$ and $\frac{p}{2}$, and they are level at $p = \frac{2}{3}$.

This is why the third row of Kagey's table for $p = \frac{2}{3}$ reads 2, 2, 2. With 3 steps the bins have probabilities $\frac{p^2}{2}$, $\frac{1-p^2}{2}$, $\frac{1-p^2}{2}$ and $\frac{p^2}{2}$, and they are level at $p = 1/\sqrt{2}$. In general there is exactly 1 way to reach a given end bin, by never turning, and very many ways to reach the middle by turning a few times, each turn costing a factor of about $1 - p$.

The answer is that the middle and the ends are level when the ball turns about $\log N$ times on average, where N is the number of steps. More precisely, the crossing value p_N of p satisfies $N(1 - p_N) = \log N - \frac{1}{2} \log \log N - 0.467 \dots + o(1)$, and we trap $N(1 - p_N)$ between 2 explicit functions of N whose distance tends to 0, for every $N \geq 2$ and not only in the limit (Theorem 3.4). For $p < p_N$ the middle is more likely, and for $p > p_N$ the ends are.

The other questions have short answers. The law of the bins is a sum over the number of turns (Theorem 2.1), with a rational generating function (Proposition 2.2); this is classical [25]. The numerators count lattice paths: rook paths for $p = \frac{2}{3}$ (OEIS A035002) and Mathar's blocked walks for $p = \frac{1}{3}$ (OEIS A348595), by the bijections of Appendix C. For each fixed $p \in (0, 1)$ the bin has a normal limit (Theorem 2.5), and Section 6 gives a local form that stays accurate as $p \rightarrow 1$ with N . On a cylinder the bin forgets the start geometrically fast and tends to the uniform law (Remark 2.7). A different way to make the board periodic closes the sequence of steps into a loop. That is the periodic Ising chain, and it moves the central crossing by about $(\log 2)/N$ in the odds $(1 - p)/p$ (Section 5). Kagey also asks about boards with more directions, such as a tetrahedron; Papers B and C of this series treat those (Section 1.4).

1.2 Setting and notation

The other papers of the series use the same notation.

The walk. The ball makes $N \geq 1$ steps. A run of the experiment is a word $w = w_1 \cdots w_N$ in the letters L (left) and R (right). The first letter is L or R with probability $\frac{1}{2}$ each. Each later letter equals the one before it with probability p and differs from it with probability $q = 1 - p$, independently of the past. When $p > 0$ we write $u = q/p$ for the odds of a switch. A *switch* of w is an index $i < N$ with $w_i \neq w_{i+1}$. We write $\text{chg}(w)$ for the number of switches and $\text{rep}(w) = N - 1 - \text{chg}(w)$ for the number of repeats, so that

$$\mathbb{P}(w) = \frac{1}{2} p^{\text{rep}(w)} q^{\text{chg}(w)}. \quad (1)$$

Bins. The ball lands in bin k if w has k letters R, for $0 \leq k \leq N$. We write $P_N(k)$ for the probability of bin k , and $P_0(0) = 1$. Bins 0 and N are the *endpoints*. Only the constant word $L \cdots L$ reaches bin 0, and only $R \cdots R$ reaches bin N , so

$$P_N(0) = P_N(N) = \frac{1}{2} p^{N-1} \quad (N \geq 1).$$

A middle bin is $m_N = \lfloor N/2 \rfloor$. For odd N the 2 middle bins have equal probability by symmetry. Kagey's row n has n bins, so it is our $N = n - 1$. For $p = \alpha/(\alpha + \beta)$ with integers $\alpha, \beta \geq 0$, not both 0, his numerator of bin k is $n_N(k) = 2(\alpha + \beta)^{N-1} P_N(k)$ (Remark 2.3).

Spins. Put $\sigma_i = +1$ if $w_i = R$ and $\sigma_i = -1$ if $w_i = L$, and let $X_N = \sigma_1 + \cdots + \sigma_N$ be the final position, so that the bin is $K_N = (N + X_N)/2$. By (1) the spins have law proportional to $\exp(J \sum_{i < N} \sigma_i \sigma_{i+1})$ with $e^{-2J} = u$: this is the one-dimensional Ising chain in zero field with free boundaries. The coupling J is positive exactly when $p > \frac{1}{2}$.

Scales. We write $L = \log N$ and $T = Nq$. The walk makes $N - 1$ choices to repeat or switch, so the expected number of switches is $(N - 1)q = T - q$. We call T the *switching scale*. All the crossings in this paper happen at T of order L or smaller.

Ratios and crossings. For a bin k with $1 \leq k \leq N - 1$ we write $a = k$ and $b = N - k$ for its numbers of letters R and L. For $p > 0$ the ratio of bin k to an endpoint depends on p only through u , and it is a polynomial in u with nonnegative integer coefficients ((13) below). We write

$$S_{a,b}(u) = \frac{P_N(a)}{P_N(0)}, \quad S_N(u) = S_{m_N, N-m_N}(u)$$

for this ratio and for its value at the middle. *The crossing* of bin k is the value of p at which bin k is exactly as likely as an endpoint, that is, $S_{a,b}(u) = 1$. It is unique (Section 4), and we write $p_{a,b}$ for it and $u_{a,b} = (1 - p_{a,b})/p_{a,b}$ for its odds. The *central crossing* is $p_N = p_{m_N, N-m_N}$, with odds u_N , and

$$z_N = N(1 - p_N) = \frac{Nu_N}{1 + u_N} \quad (2)$$

is the value of T there.

Special functions and asymptotic notation. I_0 and I_1 are the modified Bessel functions of the first kind. We use them through their power series and their behavior at infinity [28, Eqs. 10.25.2, 10.40.1],

$$I_0(2x) = \sum_{r \geq 0} \frac{x^{2r}}{(r!)^2}, \quad I_1(2x) = \sum_{r \geq 0} \frac{x^{2r+1}}{r!(r+1)!}, \quad I_\nu(z) = \frac{e^z}{\sqrt{2\pi z}} \left(1 - \frac{4\nu^2 - 1}{8z} + O(z^{-2}) \right)$$

as $z \rightarrow \infty$, for $\nu = 0, 1$. We write $f \sim g$ if $f/g \rightarrow 1$. Each $O(\cdot)$ comes with the parameters it is uniform in, and $O_a(\cdot)$ means that the constant may depend on a .

1.3 Main results

A Bessel bound at every N . Put $h = N/2$ and $F(x) = 2x(I_0(2x) + I_1(2x))$. Theorem 3.3 says that for every $N \geq 2$ and every $x > 0$,

$$0 \leq 1 - \frac{hS_N(x/h)}{F(x)} \leq \frac{x(x+1)}{h}.$$

In words, the middle ratio at odds $u = x/h$ is a Bessel function up to a relative error of at most $x(x+1)/h$. Near the crossing $x = Nu/2$ is about $L/2$, so the error is $O(L^2/N)$. Theorem 4.3 gives the same kind of bound for every bin, with relative error at most $Nu^2/2 + u$.

The central crossing at every N . Let ζ_N be the positive root of $8\zeta e^{2\zeta} = \pi N^2$, that is, $\zeta_N = \frac{1}{2}W_0(\pi N^2/4)$ with W_0 the principal branch of Lambert's function, and let $\varepsilon_N = \zeta_N(\zeta_N + 1)/N + 1/(16\zeta_N^2)$. Theorem 3.4 says that

$$\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z_N < \zeta_N + \frac{1}{8\zeta_N} \quad \text{for every } N \geq 2.$$

The width ε_N tends to 0. The proof is analytic for $N \geq 175$. For $2 \leq N \leq 174$ it rests on exact integer certificates for the sign of $S_N - 1$ at 2 points, one on each side of the root, which the Lean

kernel also checks. Two consequences are easier to read. With $c_0 = \frac{1}{2} \log(\pi/8) = -0.467\dots$, the crossing satisfies

$$z_N + \frac{1}{2} \log z_N = L + c_0 + \frac{1}{8z_N} + R_N, \quad |R_N| \leq \frac{1}{4L^2} + \frac{L^2 + L}{N} \quad (N \geq 2)$$

(Proposition 3.8), and as $N \rightarrow \infty$

$$z_N = L - \frac{1}{2} \log L + c_0 + \frac{\frac{1}{4} \log L - \frac{1}{2} c_0 + \frac{1}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right)$$

(Corollary 3.7). The convergence is slow: z_N/L is still only 0.84 at $N = 10^4$ (Figure 3).

Every bin. Let bin a be at distance a from the nearer edge, with $b = N - a \geq a$. Next to the edge the crossing is at $p_N^{\text{adj}} = (1 + \sqrt{N-1})/(2 + \sqrt{N-1})$, and at the central crossing these 2 bins are the least likely of all (Proposition 4.2). As $a \rightarrow \infty$, Theorem 4.7 expands $2\sqrt{ab}u_{a,b}$ to the same order as z_N , uniformly in $b \geq a$. For fixed a the crossing comes instead from a Laguerre polynomial: $u_{a,b} = c_a/\sqrt{b} - 1/b + O_a(b^{-3/2})$, where c_a^2 is the positive root of an explicit polynomial of degree a (Theorem 4.8). The 2 regimes match (Theorem 4.9).

Periodic boundaries. If the steps are closed into a loop, so that the last step also interacts with the first, the central crossing moves by $(\log 2)/N$ in the odds to leading order (Theorem 5.2). Under both boundaries, at odds $u_N + t/N$ the ratio of bin $N/2 + yN/\sqrt{\log N}$ to an endpoint tends to e^{t-2y^2} (Theorem 5.3).

A uniform local law. We approximate $P_N(k)$ itself by a tilted symmetric Skellam kernel, with an explicit relative error bound for every $0 < q < 1$ (Theorem 6.2) that tends to 0 uniformly in $q \leq \frac{1}{2}$ away from the edges (Corollary 6.3). This covers the diffusive regime (q fixed) and the ballistic regime (T bounded) at once.

Sections 2 to 6 prove these results in this order, Section 7 lists what was computed and what is in Lean, and the appendices hold the longer proofs and Kagey's tables.

1.4 The series

This paper is Paper A, part I of the series *The middle and the ends*. Papers B, C and D are parts II, III and IV [2, 3, 4]. The 4 papers share the following notation.

- Kagey's board: N steps (Kagey's row n has $N = n - 1$). The first step is left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p . Also $q = 1 - p$ and $u = q/p$, the odds of a switch.
- $P_N(k)$ is the probability of k right steps (the ball lands in bin k). The endpoints have $P_N(0) = P_N(N) = \frac{1}{2}p^{N-1}$.
- $L = \log N$ and $T = Nq$, the expected number of switches up to the first step. The central crossing is p_N , and $z_N = N(1 - p_N)$.
- *The crossing* is the value of p (or T) where a bin is exactly as likely as an endpoint.
- With d states (Papers B and C), a *face* F is a nonempty set of states and the *vertices* are the faces of size 1. For a chain with transition matrix $I + (T/N)Q$, λ_F is the principal Dirichlet eigenvalue of $-Q$ on F and K_F is the second-order constant of F .

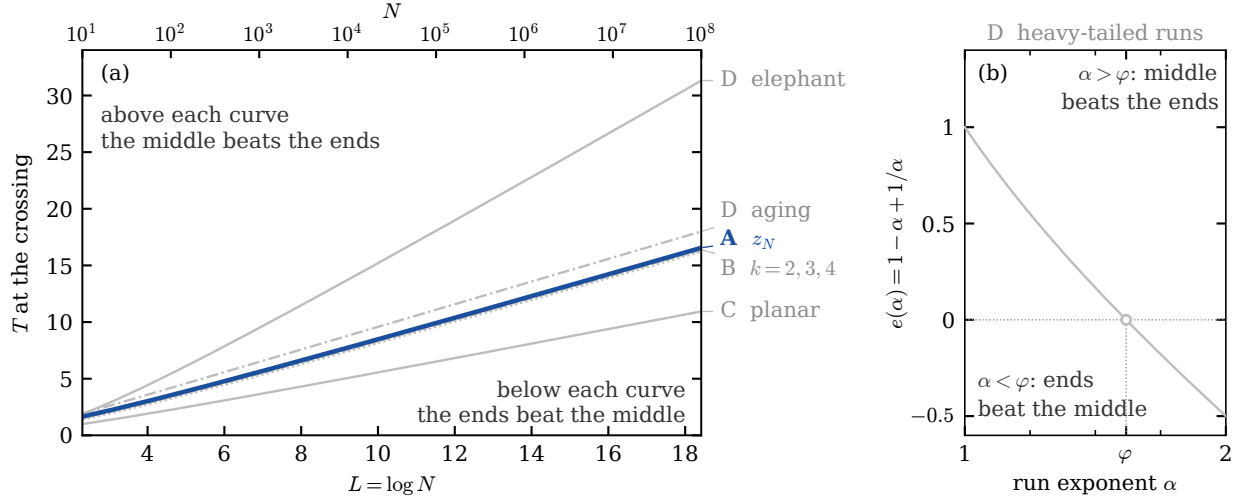


Figure 1: (a) The crossing on each model's switching scale against $L = \log N$ for Papers A to D. The scale is T for Papers A to C, $x = N\varepsilon$ for the elephant walk and the expected number of switches for the aging walk of Paper D. The bold curve is the 2-state crossing z_N of this paper, and the curves B are the crossings of Paper B for faces of $k = 2, 3, 4$ states. (b) The exponent $e(\alpha) = 1 - \alpha + 1/\alpha$ of Paper D for runs with heavy tails, which changes sign at the golden ratio φ .

This paper treats Kagey's board itself, with 2 directions, and the others use its exact law, its Bessel bounds and its crossing z_N . Paper B lets the ball move in d directions and switch to each other direction at the same rate (the d -state Potts chain); there every face ties with the vertices at first order, and Paper B resolves the tie at second order. Paper C treats a general chain with transition matrix $I + (T/N)Q$ and $T = \tau L$, where to first order the mode sits on a face that minimizes $\tau\lambda_F + |F| - 1$; the constant c_0 of this paper is one of its corollaries. Paper D asks what memory changes, and finds for the elephant random walk a crossing about twice z_N . Figure 1 shows the crossings of all 4 papers.

1.5 Related work

The correlated random walk goes back to Goldstein [16] and Gillis [15]. Renshaw and Henderson [25] study the same symmetric model, motivated by a Galton machine. They give the exact law from run counts, the generating function, the normal limit and the upturn at the endpoints, and in their Section 4 a Bessel limit at a fixed persistence scale, followed by a limit as that scale grows. Since the 2 limits are taken one after the other, there is no error bound at finite N when the scale grows like $\log N$, which is where the crossing happens. Konno [19] writes the exact law with the hypergeometric function ${}_2F_1$ and proves the telegraph limit when $p = 1 - \theta/N$ with θ fixed, again a fixed scale. Herrmann and Vallois [17] give the telegraph limit with its Bessel density, and Dekking and Kong [12] study the Markov binomial law, give its variance and find when it has 1, 2 or 3 modes. The OEIS entries [22, 23] already record both of Kagey's tables.

For the Ising chain, Antal, Droz and Rácz [5] give the magnetization law at a fixed scale T under several boundary conditions, and Dantchev and Rudnick [11] give exact free and periodic coefficients. Stepanyan et al. [26, Section III.B] study the periodic endpoint-adjacent and endpoint-center crossings directly, so the crossing question itself is not new. What we add are bounds at finite N when T is of order $\log N$, also near the edges. The Markov binomial estimates of

Čekanavičius and Mikalauskas [7, 8] assume ranges of the transition probabilities that exclude $p \rightarrow 1$. Section 6 discusses the work on local limits.

2 The exact law

This section answers Kagey's questions about the law of the bins, his numerator tables and the normal limit. The results are classical in substance [25, 12], and we prove them because the later sections use the count of words by their number of switches. Recall (1): a word with j switches has probability $\frac{1}{2}p^{N-1-j}q^j$. We also write $\rho = 2p - 1 = p - q$, the correlation $\mathbb{E}\sigma_i\sigma_{i+1}$ of 2 consecutive steps.

Theorem 2.1 (bin probabilities). *Let $N \geq 1$. Then $P_N(0) = P_N(N) = \frac{1}{2}p^{N-1}$, and for $1 \leq k \leq N - 1$*

$$P_N(k) = \frac{1}{2} \sum_{j=1}^{N-1} W(N, k, j) p^{N-1-j} q^j,$$

where, writing $a = k$ and $b = N - k$,

$$W(N, k, 2i - 1) = 2 \binom{a-1}{i-1} \binom{b-1}{i-1}, \quad W(N, k, 2i) = \binom{a-1}{i} \binom{b-1}{i-1} + \binom{a-1}{i-1} \binom{b-1}{i}$$

is the number of words with a letters R, b letters L and exactly j switches.

In words, we sort the words in bin k by their number of switches, and count each class by its run lengths.

Proof. By (1) it suffices to count words. A word with j switches consists of $j + 1$ maximal runs, alternately of R's and of L's. If $k \in \{0, N\}$ there is 1 word and it has no switch. Let $a, b \geq 1$. If $j + 1 = 2i$ there are i runs of each letter and the first letter is free. The run lengths form a composition of a into i parts and a composition of b into i parts, and there are $\binom{a-1}{i-1} \binom{b-1}{i-1}$ such pairs. If $j + 1 = 2i + 1$ the word starts and ends with the same letter. Starting with R gives $i + 1$ runs of R and i runs of L, so $\binom{a-1}{i} \binom{b-1}{i-1}$ words, and starting with L gives the other summand. $\square$

For instance $P_2 = (\frac{p}{2}, q, \frac{p}{2})$, $P_3(1) = \frac{1}{2}(1 - p^2)$ and $P_4(2) = q(1 - p + p^2)$. Each interior bin needs at least 1 switch, while the endpoints need none. So the endpoints win when p is close to 1, and Section 3 asks how close.

Proposition 2.2 (recurrence and generating function). *Let $P_N^R(k)$ and $P_N^L(k)$ be the probabilities of being in bin k after N steps with last step R, respectively L, and set them to 0 outside $0 \leq k \leq N$. Then $P_1^R(1) = P_1^L(0) = \frac{1}{2}$, $P_1^R(0) = P_1^L(1) = 0$, and for $N \geq 2$*

$$P_N^R(k) = p P_{N-1}^R(k-1) + q P_{N-1}^L(k-1), \quad P_N^L(k) = p P_{N-1}^L(k) + q P_{N-1}^R(k). \quad (3)$$

Consequently

$$\sum_{N \geq 0} \sum_{k=0}^N P_N(k) x^k t^N = \frac{1 - (p - \frac{1}{2})(1+x)t}{1 - p(1+x)t + (2p-1)xt^2}, \quad (4)$$

and for all $N \geq 2$ and all k , with $P_M(k) = 0$ outside $0 \leq k \leq M$,

$$P_N(k) = p(P_{N-1}(k) + P_{N-1}(k-1)) - (2p-1)P_{N-2}(k-1). \quad (5)$$

Proof. The recursion (3) follows by conditioning on the last 2 steps. Let $H^R = \sum_{N,k} P_N^R(k) x^k t^N$ and define H^L alike. By (3) and the initial values, $H^R = xt(\frac{1}{2} + pH^R + qH^L)$ and $H^L = t(\frac{1}{2} + pH^L + qH^R)$. The determinant of this linear system is $\Delta = 1 - p(1+x)t + \rho xt^2$, since $p^2 - q^2 = \rho$. Cramer's rule gives $H^R = \frac{1}{2}xt(1 - \rho t)/\Delta$ and $H^L = \frac{1}{2}t(1 - \rho xt)/\Delta$, and $1 + H^R + H^L$ simplifies to (4). Multiplying (4) by Δ and comparing the coefficients of $x^k t^N$ with $N \geq 2$ gives (5). $\square$

At $p = \frac{1}{2}$ the last term of (5) vanishes, and (5) is Pascal's rule with weights $\frac{1}{2}$.

Remark 2.3 (numerator tables). For $p = \alpha/(\alpha + \beta)$ with integers $\alpha, \beta \geq 0$, not both 0, Kagey's numerator tables are

$$n_N(k) = 2(\alpha + \beta)^{N-1} P_N(k) \quad (N \geq 1), \quad n_0(0) = 1,$$

not reduced. By (1), $n_N(k)$ is the sum of $\alpha^{\text{rep}(w)} \beta^{\text{chg}(w)}$ over the words with k letters R and $N - k$ letters L, so it is an integer. By (5), $n_N(k) = \alpha(n_{N-1}(k) + n_{N-1}(k-1)) - (\alpha^2 - \beta^2)n_{N-2}(k-1)$ for $N \geq 3$. Read a word as a monotone lattice path to $(k, N - k)$. Then $n_N(k)$ counts these paths with each straight continuation colored in 1 of α ways and each corner in 1 of β ways. For $p = \frac{2}{3}$ the table counts rook paths (A035002 read by antidiagonals), and for $p = \frac{1}{3}$ it counts Mathar's blocked walks (A348595). The OEIS entries already record the generating functions (by Kruchinin, Somos and Mathar) [22, 23], and our bijections explain them. Appendix C has the proofs.

2.1 Variance and the normal limit

Recall the spins σ_i , the position $X_N = \sigma_1 + \dots + \sigma_N$ and the bin $K_N = (N + X_N)/2$. The step rule says that $\sigma_{i+1} = \sigma_i s_i$, where $s_1, s_2, \dots$ are independent of each other and of σ_1 , with $\mathbb{P}(s_i = 1) = p$ and $\mathbb{P}(s_i = -1) = q$. So $\mathbb{E}s_i = \rho$.

Proposition 2.4 (mean and variance). $\mathbb{E}K_N = N/2$, and for $p < 1$

$$\text{Var } K_N = \frac{1}{4} \left(N \frac{1 + \rho}{1 - \rho} - \frac{2\rho(1 - \rho^N)}{(1 - \rho)^2} \right) = \frac{Np}{4(1 - p)} - \frac{(2p - 1)(1 - (2p - 1)^N)}{8(1 - p)^2}.$$

For $p = 1$, $\text{Var } K_N = N^2/4$.

Proof. Note that $\mathbb{E}\sigma_1 = 0$ and $\sigma_{i+m} = \sigma_i s_i \dots s_{i+m-1}$, so $\mathbb{E}\sigma_i = 0$ and $\mathbb{E}\sigma_i \sigma_{i+m} = \rho^m$. Hence $\text{Var } X_N = N + 2 \sum_{m=1}^{N-1} (N - m) \rho^m$. Summing the 2 geometric series gives the first form, and the second follows from $(1 + \rho)/(1 - \rho) = p/q$ and $1 - \rho = 2q$. $\square$

Theorem 2.5 (normal limit). Let $0 < p < 1$ be fixed. Then $(K_N - N/2)/\sqrt{N}$ converges in distribution, as $N \rightarrow \infty$, to the normal law with mean 0 and variance $p/(4(1 - p))$.

So persistence keeps the bell shape but widens it by the factor $\sqrt{p/q}$ compared with the ordinary board. The idea of the proof is that the characteristic function is a power of a 2×2 transfer matrix, whose largest eigenvalue near $\theta = 0$ is $1 - \frac{p}{2q}\theta^2 + O(\theta^4)$.

Proof. Let $\varphi_N(\theta) = \mathbb{E}e^{i\theta X_N}$, and let v_N be the vector with entries $\mathbb{E}[e^{i\theta X_N}; \sigma_N = \pm 1]$. By (3), $v_{N+1} = Mv_N$ with

$$M = \begin{pmatrix} pe^{i\theta} & qe^{i\theta} \\ qe^{-i\theta} & pe^{-i\theta} \end{pmatrix}, \quad \text{tr } M = 2p \cos \theta, \quad \det M = \rho.$$

The eigenvalues are $\lambda_{\pm}(\theta) = p \cos \theta \pm \sqrt{q^2 - p^2 \sin^2 \theta}$. For $|\sin \theta| < q/p$ they are real and distinct, so $\varphi_N = c_+ \lambda_+^{N-1} + c_- \lambda_-^{N-1}$ for $N \geq 1$. The values $\varphi_1 = \cos \theta$ and $\varphi_2 = p \cos 2\theta + q$ give $c_+ = (\varphi_2 - \lambda_- \cos \theta)/(\lambda_+ - \lambda_-)$ and $c_- = \cos \theta - c_+$. As $\theta \rightarrow 0$ we have $\lambda_- \rightarrow \rho$, $c_+ \rightarrow 1$, $c_- \rightarrow 0$ and

$$\lambda_+(\theta) = p(1 - \frac{\theta^2}{2}) + q(1 - \frac{p^2 \theta^2}{2q^2}) + O(\theta^4) = 1 - \frac{p}{2q} \theta^2 + O(\theta^4).$$

Put $\theta = s/\sqrt{N}$ with s fixed. Since $|\rho| < 1$, we get $\varphi_N(s/\sqrt{N}) \rightarrow \exp(-\frac{p}{2q}s^2)$. By Lévy's continuity theorem [6, Theorem 26.3], $X_N/\sqrt{N}$ converges to the normal law with variance p/q . Finally $K_N - N/2 = X_N/2$. $\square$

For $p = 1$ the walk ends in bin 0 or bin N , and for $p = 0$ in a middle bin, so these 2 cases have no normal limit.

Remark 2.6 (zeros of the generating polynomial). Let $N \geq 2$ and $g_N(x) = \sum_k P_N(k)x^k$. The zeros of g_N behave differently on the 2 sides of $p = \frac{1}{2}$. First let $0 < p \leq \frac{1}{2}$. By (5), $g_N = p(1+x)g_{N-1} + (1-2p)xg_{N-2}$ with a nonnegative last coefficient. Then every g_N has only real zeros, as in the proof of [12, Proposition 4.2], and they are negative since g_N has positive coefficients. By a classical result recalled by Pitman [24, Proposition 1], K_N then has the law of a sum of N independent Bernoulli variables. So each row is log-concave and no bin is less likely than the endpoints. This fits with Section 4, where every endpoint crossing is at least $\frac{2}{3}$.

Now let $\frac{1}{2} \leq p < 1$, so that $u \leq 1$. Put $x_{i,i+1} = x_{i+1,i} = u$ for $1 \leq i < N$ and $x_{ij} = 1$ for the other pairs $i \neq j$. For $S \subseteq \{1, \dots, N\}$ let w be the word with R exactly at the positions in S . Each switch of w is a pair of neighbors split by S , so $\prod_{i \in S, j \notin S} x_{ij} = u^{\text{chg}(w)}$, and (1) gives

$$g_N(x) = \frac{p^{N-1}}{2} \sum_{S \subseteq \{1, \dots, N\}} x^{|S|} \prod_{i \in S, j \notin S} x_{ij}.$$

The x_{ij} are real and symmetric with $|x_{ij}| \leq 1$, so by the Lee–Yang circle theorem [20], in the form quoted by Ghosh [14], every zero of g_N lies on the circle $|x| = 1$. It applies because the boundaries are free and the first step is uniform, so there is no boundary field, and because $u \leq 1$ (that is, $J \geq 0$).

At $p = \frac{1}{2}$ we have $g_N = ((1+x)/2)^N$, and both statements hold. For $\frac{1}{2} < p < 1$ the zeros are never all real. Suppose they were. Then K_N would again be a sum of N independent Bernoulli variables, with variance at most $N/4$. But the proof of Proposition 2.4 gives $\text{Var } X_N = N + 2 \sum_{m=1}^{N-1} (N-m)\rho^m > N$, so $\text{Var } K_N > N/4$.

Remark 2.7 (cylinder). Kagey also asks about a board wrapped into a cylinder. Suppose each row has c pegs around the cylinder and the local rule is unchanged. Then the position is X_N modulo $2c$ in half-peg units. The Fourier coefficients of its law are $\varphi_N(\pi l/c)$, $0 \leq l < 2c$, and they equal 1 and $(-1)^N$ at $l = 0$ and $l = c$. Let $0 < p < 1$ and $\theta \notin \pi\mathbb{Z}$. Then both eigenvalues of M have modulus less than 1. Real ones satisfy $|\lambda_{\pm}| \leq p|\cos \theta| + q < 1$, and complex ones have modulus $\sqrt{\rho} < 1$. So $\varphi_N(\theta) \rightarrow 0$ geometrically. This also holds for a double eigenvalue $\lambda = p \cos \theta$, since then $(M - \lambda I)^2 = 0$ by the Cayley–Hamilton theorem and $M^n = \lambda^n I + n\lambda^{n-1}(M - \lambda I)$. Hence the law tends geometrically fast to the uniform law on the c positions of the correct parity. In words, on a cylinder the ball forgets where it started, and there is no normal shape in the limit.

3 The central crossing at every N

We now turn to Kagey's main question. Let $N \geq 2$, let $m_N = \lfloor N/2 \rfloor$ and let $p > 0$. By Theorem 2.1 the ratio of the middle bin to an endpoint is

$$\frac{P_N(m_N)}{P_N(0)} = S_N(u) := \sum_{j=1}^{N-1} W(N, m_N, j) u^j, \quad (6)$$

a polynomial in u with nonnegative integer coefficients and $W(N, m_N, 1) = 2$. The middle and the endpoints are level exactly when $S_N(u) = 1$.

Here is why this happens when T is near $\log N$. Let $a = m_N$ and $b = N - m_N$, both about $N/2$. A word in the middle bin with $2r + 1$ switches has $r + 1$ runs of each letter, and there are $2 \binom{a-1}{r} \binom{b-1}{r} \approx 2(N/2)^{2r} / (r!)^2$ of them when r is small compared with $\sqrt{N}$. So these words contribute about $2u \sum_r (Nu/2)^{2r} / (r!)^2 = 2u I_0(Nu)$ to $S_N(u)$, and in the same way the words with an even number of switches contribute about $2u I_1(Nu)$. For large $w = Nu$ we have $I_0(w) + I_1(w) \approx 2e^w / \sqrt{2\pi w}$, so $S_N(u) \approx 4ue^w / (N\sqrt{2\pi w})$. Setting this equal to 1 gives $w + \frac{1}{2} \log w \approx L + \frac{1}{2} \log(\pi/8)$. So at the crossing w , and with it T , is close to $L - \frac{1}{2} \log L + c_0$ with $c_0 = \frac{1}{2} \log(\pi/8)$. The rest of this section makes this precise at every N . First we show that the crossing exists and is unique.

Theorem 3.1. (a) For every $N \geq 2$ there is exactly one $p = p_N \in [0, 1]$ with $P_N(m_N) = P_N(0)$, and $p_N \geq \frac{2}{3}$. The middle is more likely than the endpoints for $p < p_N$ and less likely for $p > p_N$.
(b) $p_2 = \frac{2}{3}$, $p_3 = 1/\sqrt{2}$, p_4 is the real root of $3p^3 - 4p^2 + 4p - 2$, and $p_N < p_{N+1}$ for all $N \geq 2$.
(c) $p_N \rightarrow 1$ as $N \rightarrow \infty$, and more precisely $1 - p_N \sim (\log N)/N$.

In words: there is a single switch-over point, it moves toward $p = 1$ as the board grows, and $N(1 - p_N)$ grows like $\log N$.

Proof. (a) At $p = 0$ the endpoints have probability 0 and the middle does not. For $p > 0$ the equation is $S_N(u) = 1$ by (6). The polynomial S_N is continuous and strictly increasing on $[0, \infty)$ with $S_N(0) = 0$ and $S_N(u) \geq 2u$, so there is exactly one root $u_N \in (0, \frac{1}{2}]$. The map $u \mapsto p = 1/(1 + u)$ is a decreasing bijection from $[0, \infty)$ to $(0, 1]$. Hence $p_N = 1/(1 + u_N) \geq \frac{2}{3}$, and the middle wins exactly when $u > u_N$.

(b) $S_2 = 2u$, $S_3 = 2u + u^2$ and $S_4 = 2u + 2u^2 + 2u^3$ give the 3 values. For p_4 , substitute $u = (1 - p)/p$ in $S_4(u) = 1$. For the monotonicity we compare coefficients. Going from N to $N + 1$, one entry of the middle pair $(a, b) = (\lfloor N/2 \rfloor, \lceil N/2 \rceil)$ grows by 1 and the other is unchanged. The formulas of Theorem 2.1 are symmetric in (a, b) , and every $\binom{a-1}{r}$ is nondecreasing in a . So $W(N + 1, m_{N+1}, j) \geq W(N, m_N, j)$ for every j . The coefficient of u^2 is $a + b - 2 = N - 2$, which increases strictly. Hence $S_{N+1}(u) > S_N(u)$ for $u > 0$, so $u_{N+1} < u_N$.

(c) By Corollary 3.7(b) below, used only for $N \geq 175$ where its proof is analytic, $z_N = L + O(\log L)$, which gives (c). There is no circularity: the proof of Corollary 3.7 uses Theorem 3.4, whose proof uses only part (i) of Theorem 3.3. $\square$

In Kagey's terms, the middle and the endpoints of row $N + 1$ are level at $p = p_N$. The least value is $p_2 = \frac{2}{3}$ for the odd rows and $p_3 = 1/\sqrt{2}$ for the even rows after the second (which is 1, 1 for every p). There is no greatest value, since $p_N \uparrow 1$. The tenth row has $p_9 = 0.82829\dots$, just below $\frac{5}{6}$, so its middle and endpoints are nearly level in his third histogram (Figure 2 and Table 1).

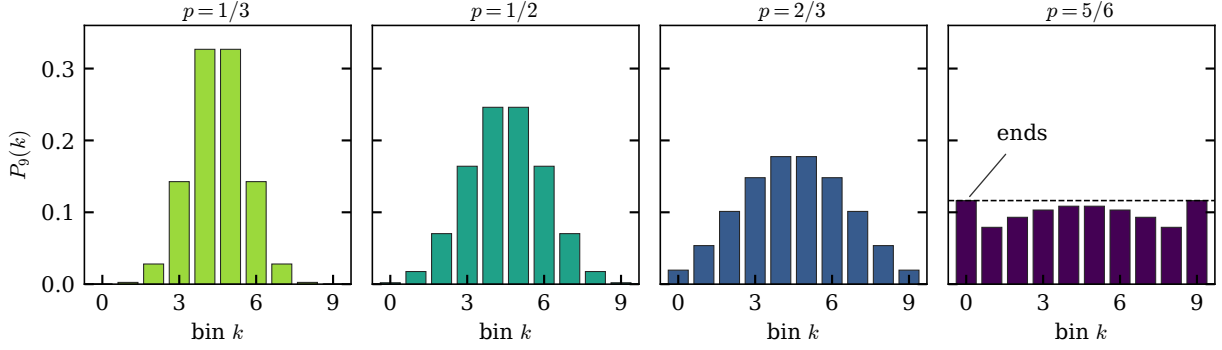


Figure 2: The bin probabilities $P_9(k)$ of Kagey's tenth row for $p = \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{5}{6}$. At $p = \frac{5}{6}$, just above p_9 , the 2 middle bins are slightly below the endpoints (dashed line).

N	p_N	N	p_N	N	p_N	N	p_N
2	0.66666667	6	0.78871211	10	0.83840411	50	0.94200405
3	0.70710678	7	0.80361731	11	0.84652725	100	0.96496633
4	0.74487450	8	0.81759938	12	0.85426060	200	0.97939018
5	0.76759188	9	0.82829163	20	0.89293837	400	0.98812464

Table 1: The value p_N at which the middle and endpoint bins of row $N + 1$ are equally likely.

3.1 A Bessel bound at every N

The heuristic above replaced the number $\binom{a-1}{r}$ of ways to split a letters into $r + 1$ runs by $a^r/r!$. After this change the run count is exactly the power series of I_0 and I_1 . The next lemma controls the cost of the change.

Lemma 3.2 (clipped products). (a) If $t_1, \dots, t_r \geq 0$ and $(v)_+ = \max(v, 0)$, then $0 \leq 1 - \prod_{k=1}^r (1 - t_k)_+ \leq \sum_{k=1}^r t_k$. (b) For integers $m \geq 1$ and $r \geq 0$, $U_r(m) := \frac{r!}{m^r} \binom{m-1}{r} = \prod_{k=1}^r (1 - \frac{k}{m})_+$, so $0 \leq 1 - U_r(m) \leq \frac{r(r+1)}{2m}$. (c) For $x \geq 0$,

$$\sum_{r \geq 0} \frac{r(r+1)x^{2r}}{(r!)^2} = x^2 I_0(2x) + x I_1(2x),$$

$$\sum_{r \geq 0} \frac{r(r+1)x^{2r+1}}{r!(r+1)!} = x^2 I_1(2x), \quad \sum_{r \geq 0} \frac{(r+1)x^{2r+1}}{r!(r+1)!} = x I_0(2x).$$

Proof. (a) Every factor lies in $[0, 1]$, so the product is at most 1. If some $t_k \geq 1$ the product is 0 and the sum is at least 1. Otherwise $\prod_k (1 - t_k) \geq 1 - \sum_k t_k$ by induction on r . (b) Note that $\frac{r!}{m^r} \binom{m-1}{r} = \prod_{k=1}^r \frac{m-k}{m}$. For $r < m$ every factor is positive, and for $r \geq m$ both sides vanish, the right side through the factor $k = m$. The bound is (a) with $t_k = k/m$. (c) Use the series of $I_0(2x)$ and $I_1(2x)$ from Section 1.2. Write $r(r+1) = r^2 + r$ in the first sum, cancel the factors r and $r+1$ against the factorials, and shift the index. $\square$

Theorem 3.3 (uniform Bessel approximation). Put $h = N/2$ and $F(x) = 2x(I_0(2x) + I_1(2x))$, and recall that $u_N = (1 - p_N)/p_N$.

(i) For every $N \geq 2$ and $x > 0$,

$$0 \leq 1 - \frac{hS_N(x/h)}{F(x)} \leq \frac{x(x+1)}{h}. \quad (7)$$

(ii) Let $x_N = hu_N$, let x_B be the unique positive root of $F(x_B) = h$, and put $\delta_N = x_N(x_N + 1)/h$.

Then $x_B \leq x_N$, and $x_N - x_B \leq -\frac{1}{2} \log(1 - \delta_N)$ whenever $\delta_N < 1$.

(iii) With $u_B = x_B/h$ we have $0 \leq u_N - u_B = O(L^2/N^2)$ as $N \rightarrow \infty$.

In words, hS_N never exceeds the Bessel function F , and falls short of it by at most the fraction $x(x+1)/h$. At the crossing x is about $L/2$, so the root u_B of $F = h$ is within a relative $O(L^2/N)$ of u_N . Since (i) holds at every N , it lets us locate p_N at every N below.

Proof. (i) *Step 1: the clipped factors.* Let $a = \lfloor N/2 \rfloor$ and $b = \lceil N/2 \rceil$. As in Lemma 3.2(b), $A_r = \frac{r!}{h^r} \binom{a-1}{r} = \prod_{k=1}^r (1 - \frac{h-a+k}{h})_+$, and B_r is the same with b in place of a . Since $a+b=2h$ and $h-a \in \{0, \frac{1}{2}\}$, the $2r$ numbers $(h-a+k)/h$ and $(h-b+k)/h$ are nonnegative with sum $r(r+1)/h$, so Lemma 3.2(a) gives $0 \leq 1 - A_r B_r \leq r(r+1)/h$. In the same way the deficits of $A_{r+1} B_r$ and $A_r B_{r+1}$ are at most $((r+1)^2 + h - a)/h$ and $((r+1)^2 + h - b)/h$. Averaging cancels the parity offsets, so $0 \leq 1 - \frac{1}{2}(A_{r+1} B_r + A_r B_{r+1}) \leq (r+1)^2/h$.

Step 2: summing. With $u = x/h$ the run-count formula of Theorem 2.1 becomes

$$S_N(u) = 2u \left[\sum_{r \geq 0} \frac{x^{2r}}{(r!)^2} A_r B_r + \sum_{r \geq 0} \frac{x^{2r+1}}{r!(r+1)!} \frac{A_{r+1} B_r + A_r B_{r+1}}{2} \right].$$

Without the factors A_r, B_r the bracket is $I_0(2x) + I_1(2x)$ and the right side is $F(x)/h$. By Step 1 and Lemma 3.2(c), $F(x)/h - S_N(u)$ lies between 0 and $2u/h$ times $\sum_r r(r+1) \frac{x^{2r}}{(r!)^2} + \sum_r (r+1)^2 \frac{x^{2r+1}}{r!(r+1)!} = x(x+1)(I_0(2x) + I_1(2x))$. Dividing by $F(x)/h = 2u(I_0(2x) + I_1(2x))$ proves (i). This covers every r , also those beyond the degree of the polynomial, where the clipped products vanish and Lemma 3.2(a) still applies.

(ii) Differentiating the series term by term gives $F' = 2F + 2I_0(2x)$. So F increases from 0 to ∞ , x_B is unique, and $(\log F)' \geq 2$. At $x = x_N$ we have $hS_N(u_N) = h$, so (i) gives $h \leq F(x_N)$, hence $x_B \leq x_N$, and $F(x_N) \leq h/(1 - \delta_N)$ when $\delta_N < 1$. Integrating $(\log F)' \geq 2$ from x_B to x_N gives $2(x_N - x_B) \leq \log(F(x_N)/h) \leq -\log(1 - \delta_N)$.

(iii) By Theorem 3.1(c), whose proof does not use (iii), $x_N \sim L/2$. So $\delta_N = O(L^2/N)$, and (ii) gives $x_N - x_B = O(L^2/N)$. Divide by h . $\square$

3.2 The crossing at every N

We measure the crossing by $z_N = N(1 - p_N) = Nu_N/(1 + u_N)$, the value of T at $p = p_N$. Let $\psi(z) = z + \frac{1}{2} \log z$ and $c_0 = \frac{1}{2} \log(\pi/8) = -0.46735\dots$, and let ζ_N be the positive root of

$$\psi(\zeta_N) = L + c_0, \quad \text{that is,} \quad 8\zeta_N e^{2\zeta_N} = \pi N^2 \quad \text{or} \quad \zeta_N = \frac{1}{2} W_0(\pi N^2/4), \quad (8)$$

with W_0 the principal branch of the Lambert function. This is the heuristic equation from the start of the section with w replaced by ζ . Since $\zeta e^{2\zeta}$ increases, ζ_N increases with N . Also $\zeta_N \leq L$, since $\psi(L) \geq L + c_0$ when $L \geq \pi/8$. Finally let $\varepsilon_N = \zeta_N(\zeta_N + 1)/N + 1/(16\zeta_N^2)$.

Theorem 3.4 (the crossing at every N). *For every $N \geq 2$,*

$$\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z_N < \zeta_N + \frac{1}{8\zeta_N}.$$

In words, z_N lies in an explicit interval of width ε_N just below $\zeta_N + 1/(8\zeta_N)$. Since $\zeta_N \leq L$ and $\zeta_N \rightarrow \infty$, the width tends to 0. It is 0.440, 0.163, 0.0387 and 0.00778 at $N = 10, 10^2, 10^3, 10^4$ (Table 2).

Here is where the 2 terms ζ_N and $1/(8\zeta_N)$ come from. Let $\Phi(w) = F(w/2) = w(I_0(w) + I_1(w))$ and $E(z) = \sqrt{\pi z/2} e^{-z}(I_0(z) + I_1(z))$, which tends to 1 as $z \rightarrow \infty$. Put $w_N = Nu_N$ and $\Lambda_N = 2\Phi(w_N)/N$. At the crossing, (7) and (9) below give $\psi(w_N) + \log E(w_N) = L + c_0 + \log \Lambda_N$ with $0 \leq \log \Lambda_N \leq -\log(1 - w_N(w_N + 2)/(2N))$. With $E = \Lambda_N = 1$ this would say $w_N = \zeta_N$. Since $\log E(w) = -1/(8w) + O(w^{-2})$ and $z_N = w_N + O(w_N^2/N)$, it suggests $z_N = \zeta_N + 1/(8\zeta_N) + O(\zeta_N^{-2} + \zeta_N^2/N)$. The theorem makes this error explicit and one-sided. The proof needs 2 lemmas: the first turns a comparison of z_N with a test value into a sign condition, and the second bounds E with explicit constants.

Lemma 3.5 (root bracket). *Let $N \geq 2$ and $w > 0$. Then*

$$\log \Phi(w) - \log \frac{N}{2} = \psi(w) - \psi(\zeta_N) + \log E(w). \quad (9)$$

(a) *If $\Phi(w) \leq N/2$, then $Nw/(N+w) \leq z_N$, strictly if $\Phi(w) < N/2$. (b) If $\Phi(w)(1 - \frac{w(w+2)}{2N}) > N/2$, then $z_N < Nw/(N+w)$.*

Proof. Note that $\log \Phi(w) = \psi(w) + \frac{1}{2} \log \frac{2}{\pi} + \log E(w)$ and $\log \frac{N}{2} - \psi(\zeta_N) = -\log 2 - c_0 = \frac{1}{2} \log \frac{2}{\pi}$, which gives (9). Now apply (7) with $x = w/2$. Then $F(x) = \Phi(w)$ and $x(x+1)/h = w(w+2)/(2N)$, so

$$\Phi(w) \left(1 - \frac{w(w+2)}{2N}\right) \leq \frac{N}{2} S_N\left(\frac{w}{N}\right) \leq \Phi(w).$$

(a) The hypothesis gives $S_N(w/N) \leq 1 = S_N(u_N)$, strictly if $\Phi(w) < N/2$. Since S_N is strictly increasing, $w/N \leq u_N$, strictly in that case. The map $v \mapsto Nv/(1+v)$ increases and sends w/N to $Nw/(N+w)$ and u_N to z_N . (b) Now $S_N(w/N) > 1$, so $w/N > u_N$. $\square$

Lemma 3.6 (Bessel bounds). *For every $z > 0$,*

$$1 - \frac{1}{8z} - \frac{3}{128z^2} - \frac{45}{512z^3} \leq E(z) \leq 1 - \frac{1}{8z} - \frac{3}{128z^2} + \frac{15}{512z^3},$$

and $E(z) \leq 1$. In particular $E(z) \leq 1 - \frac{1}{8z}$ for $z \geq \frac{5}{4}$.

This lemma replaces the large-argument expansion of $I_0 + I_1$ by bounds with explicit errors. By [28, Eq. 10.32.3] and a substitution, $E(z) = \pi^{-1/2} \int_0^{2z} e^{-v} v^{-1/2} \sqrt{1 - v/(2z)} dv$. We bound $\sqrt{1 - y}$ above and below by cubics that share its Taylor terms up to y^2 , extend the integral to ∞ , and use $\Gamma(k + \frac{1}{2})/\Gamma(\frac{1}{2}) = 1, \frac{1}{2}, \frac{3}{4}, \frac{15}{8}$ for $k = 0, 1, 2, 3$. The details are in Appendix A.

Proof of Theorem 3.4. Step 1: the split at $N = 175$. Since $N = \sqrt{8\zeta_N/\pi} e^{\zeta_N}$ and $\sqrt{32/\pi} e^4 = 174.25\dots$, we have $\zeta_N \geq 4$ exactly when $N \geq 175$. Write $\zeta = \zeta_N$ and $U = \zeta + \frac{1}{8\zeta}$.

Step 2: the upper bound for $N \geq 175$. We apply Lemma 3.5(b) at $W = NU/(N - U)$, for which $NW/(N + W) = U$. By (9) it suffices that $G_U = \psi(W) - \psi(\zeta) + \log E(W) + \log(1 - W(W + 2)/(2N)) > 0$. Expanding each term with Lemma 3.6 gives $G_U \geq \frac{1}{32\zeta^2} + \frac{\zeta(\zeta-1)}{2N} + O(\zeta^{-3} + N^{-1} + \zeta^4 N^{-2})$, and Appendix A makes every term explicit for $\zeta \geq 4$.

Step 3: the lower bound for $N \geq 175$. We apply Lemma 3.5(a) at $W' = N\ell/(N - \ell)$ with $\ell = U - \varepsilon_N$. It suffices that $G_\ell = \psi(W') - \psi(\zeta) + \log E(W') < 0$, and we get $G_\ell \leq -\frac{3}{128\zeta^2} - \frac{\zeta}{N} + O(\zeta^{-3} + N^{-1} + \zeta^4 N^{-2})$, again with explicit constants for every $\zeta \geq 4$ in Appendix A. The term $-\zeta/N$ is why ε_N contains $\zeta(\zeta + 1)/N$.

Step 4: $2 \leq N \leq 174$, by certificates. Let $d = 10^9$ and let $c < c'$ be integers with $c' = c + 1$ ($c' = c + 2$ at $N = 2$, where $u_2 = \frac{1}{2}$ is a grid point). The sign of $S_N(c/d) - 1$ is the sign of $n_N(m_N) - d^{N-1}$, where $n_N(m_N)$ is the numerator of Proposition C.1 at $(\alpha, \beta) = (d, c)$, by the identity at the end of Appendix A. We computed these integers at c and c' by 2 exact methods, the run-count sum of Theorem 2.1 and the recursion (3) with integer weights d, c in place of p, q . They agreed for every $N \leq 174$ and gave $c/d < u_N < c'/d$, so $z(c/d) < z_N < z(c'/d)$ with $z(u) = Nu/(1+u)$. We then checked $\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N < z(c/d)$ and $z(c'/d) < \zeta_N + \frac{1}{8\zeta_N}$ in 60-digit decimal arithmetic, with ζ_N known to within 10^{-45} . We certified this bracket for ζ_N by the sign of $\psi(\zeta) - L - c_0$, which is the sign of $8\zeta e^{2\zeta} - \pi N^2$, at both of its ends, using a correctly rounded logarithm. The smallest margins are 0.033193 (upper) and 0.085591 (lower), both at $N = 174$, so every margin exceeds 0.03, far above the rounding error. The Lean proof checks the same comparisons with rigorous rational bounds on ζ_N and π . The certificates are in `paperA/data/every_n_certificates.csv`, made by `verify_every_n.py`. As a cross-check they pass for every $N \leq 1000$ (with both methods for $N \leq 400$ and the run-count sum alone above). $\square$

N	z_N	ζ_N	$z_N - \zeta_N - \frac{1}{8\zeta_N}$	$-\varepsilon_N$	R_N
2	0.666666666667	0.5368290974	-0.1030112	-0.62938	+0.050643
10	1.615958871376	1.6001732978	-0.0623310	-0.44048	-0.056660
100	3.503367368092	3.5100055489	-0.0422507	-0.16337	-0.043265
174	3.996779704730	3.9987131703	-0.0331935	-0.11878	-0.033450
175	4.001967989170	4.0038072816	-0.0330596	-0.11838	-0.033304
1000	5.590351166975	5.5807388611	-0.0127862	-0.03873	-0.011887
10^4	7.734265514999	7.7210118331	-0.0029359	-0.00778	-0.002051

Table 2: The crossing against Theorem 3.4, and the remainder R_N of (12). We computed z_N by run-count bisection in 60-digit decimal arithmetic and by a floating-point dynamic program, which agree to the digits shown.

The bracket is sharp but implicit, since it involves ζ_N . The next corollary turns it into statements about p_N and into expansions in $L = \log N$.

Corollary 3.7. *Put $\nu_N = L - \frac{1}{2} \log L + c_0$.*

(a) *For every $N \geq 2$, $1 - (\zeta_N + \frac{1}{8\zeta_N})/N < p_N < 1 - (\zeta_N + \frac{1}{8\zeta_N} - \varepsilon_N)/N$.*

(b) *For every $N \geq 2$, $\nu_N \leq \zeta_N \leq \nu_N + \frac{\log L - 2c_0}{4\nu_N}$ and*

$$-\varepsilon_N < z_N - \nu_N < \frac{\log L - 2c_0 + \frac{1}{2}}{4\nu_N}. \quad (10)$$

In particular $z_N = L - \frac{1}{2} \log L + c_0 + O(\log L/L)$ as $N \rightarrow \infty$.

(c) *As $N \rightarrow \infty$,*

$$z_N = \nu_N + \frac{\frac{1}{4} \log L - \frac{1}{2} c_0 + \frac{1}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right). \quad (11)$$

(d) *$z_N > \zeta_N$ for every $N \geq 2774$.*

In words: (a) answers Kagey's question with an explicit interval for p_N ; (b) and (c) give the first 3 and 4 terms of z_N in L , the 3-term form with bounds at every N ; and (d) says on which side of ζ_N the crossing lies for large N .

Proof. (a) is Theorem 3.4 with $p_N = 1 - z_N/N$.

(b) By (8), $\zeta_N = L + c_0 - \frac{1}{2} \log \zeta_N$. Since $\zeta_N \leq L$, this gives $\zeta_N \geq \nu_N$, and $\nu_N > 0$ since it increases in N and $\nu_2 = 0.41$. Also $\zeta_N - \nu_N = \frac{1}{2} \log(L/\zeta_N) \leq (L - \zeta_N)/(2\zeta_N)$ and $L - \zeta_N = \frac{1}{2} \log \zeta_N - c_0 \leq \frac{1}{2} \log L - c_0$. Hence $\zeta_N - \nu_N \leq (\log L - 2c_0)/(4\nu_N)$. For z_N , add Theorem 3.4 and $0 < \frac{1}{8\zeta_N} \leq \frac{1}{8\nu_N}$. The last claim follows since $\varepsilon_N = O(L^2/N + L^{-2})$.

(c) Put $\eta = \zeta_N - \nu_N = \frac{1}{2} \log(L/\zeta_N)$. By (b), $L - \zeta_N = \frac{1}{2} \log L - c_0 + O(\log L/L)$, so $\eta = \frac{L - \zeta_N}{2L} + O(\frac{(\log L)^2}{L^2}) = \frac{\frac{1}{2} \log L - \frac{1}{2} c_0}{L} + O(\frac{(\log L)^2}{L^2})$. Also $\frac{1}{8\zeta_N} = \frac{1}{8L} + O(\frac{\log L}{L^2})$ and $\varepsilon_N = O(L^2/N + L^{-2})$, and Theorem 3.4 gives (c).

(d) By Theorem 3.4, $z_N - \zeta_N > \frac{1}{8\zeta_N} - \varepsilon_N$, which is positive when $f(\zeta_N) > 1$, where $f(\zeta) = n(\zeta)(\frac{1}{8\zeta} - \frac{1}{16\zeta^2})/(\zeta(\zeta + 1))$ and $n(\zeta) = \sqrt{8\zeta/\pi} e^\zeta$, so that $n(\zeta_N) = N$. The logarithmic derivative of f is $1 + \frac{1}{2\zeta} - \frac{2}{\zeta} + \frac{1}{2\zeta^2 - \zeta} - \frac{1}{\zeta + 1} > 1 - \frac{3}{2\zeta} - \frac{1}{\zeta + 1} > 0$ for $\zeta \geq 3$. Since ζ_N increases with N , once $f(\zeta_N) > 1$ it stays above 1. This step is computer-assisted. In 60-digit decimal arithmetic, with ζ_N bracketed within 10^{-45} as in Step 4 above, we found $f(\zeta_{2774}) - 1 = 2.1 \times 10^{-4}$, and the sign holds on the whole bracket (`verify_every_n.py` and `verify_every_n_guards.py`). Since $\zeta_{2774} > 3$, this proves (d). The condition fails at $N = 2773$. $\square$

The heuristic suggested an equation for z_N of the form $\psi(z_N) \approx L + c_0 + 1/(8z_N)$. The next proposition proves it at every N , with an explicit remainder.

Proposition 3.8 (the crossing equation). *For every $N \geq 2$,*

$$z_N + \frac{1}{2} \log z_N = L + c_0 + \frac{1}{8z_N} + R_N, \quad -\frac{z_N^2 + z_N}{N} \leq R_N \leq \frac{1}{16z_N^2}. \quad (12)$$

Moreover $L/2 \leq z_N \leq L$, so $|R_N| \leq \frac{1}{4L^2} + \frac{L^2 + L}{N}$ for every $N \geq 2$.

Paper C [3] gets this equation with an error $O(1/L)$ in place of $1/(8z_N) + R_N$, as a special case of its general crossing equation, and Paper D [4] compares the elephant walk with it.

The idea of the proof is to write R_N exactly, using (7) and (9) at $w = w_N$ (formula (45)). For $N \geq 175$, Lemma 3.6 puts the Bessel part of this formula between 0 and $0.0559/z_N^2$, and the other terms lie between $-(z_N^2 + z_N)/N$ and 0. For $N \leq 174$ the proof uses the same certificates as Theorem 3.4. The details are in Appendix A.

Remark 3.9 (the sign of $z_N - \zeta_N$, computer-assisted). We have $z_N > \zeta_N$ for $N \leq 15$, $z_N < \zeta_N$ for $16 \leq N \leq 218$, and $z_N > \zeta_N$ for every $N \geq 219$. For $N \leq 1000$ the certificates decide the sign, since ζ_N never lies in $[z(c/d), z(c'/d)]$. The smallest distance is 1.66×10^{-5} , at $N = 218$. Both exact methods cover $N \leq 400$, and the run-count sum alone covers the rest. For $1001 \leq N \leq 2773$ we checked $S_N(v) < 1$ in exact integers from the run-count sum (`verify_every_n_sign.py`), at a rational v above $\zeta_N/(N - \zeta_N)$ by at least 1.2×10^{-11} , where $z(\zeta_N/(N - \zeta_N)) = \zeta_N$. These integers were cross-checked by floating-point run-count and dynamic-program sign tests for every $N \leq 3000$, and by the integer recursion at $N = 401, 1000, 1008, 2000, 2773$. For $N \geq 2774$ the sign is Corollary 3.7(d).

How good are the simpler approximations? The ratios u_B/u_N of the Bessel root of Theorem 3.3 to the true odds are 0.9761, 0.9864, 0.9924 at $N = 100, 200, 400$ and 0.99579, 0.99769, 0.99874 at $N = 800, 1600, 3200$. We predicted intervals for the last 3 before computing them (`ledger.md`), and Theorem 3.3 proves the approximation behind those predictions. Figure 3 compares the exact crossing with the expansions of Corollary 3.7 up to $N = 10^4$.

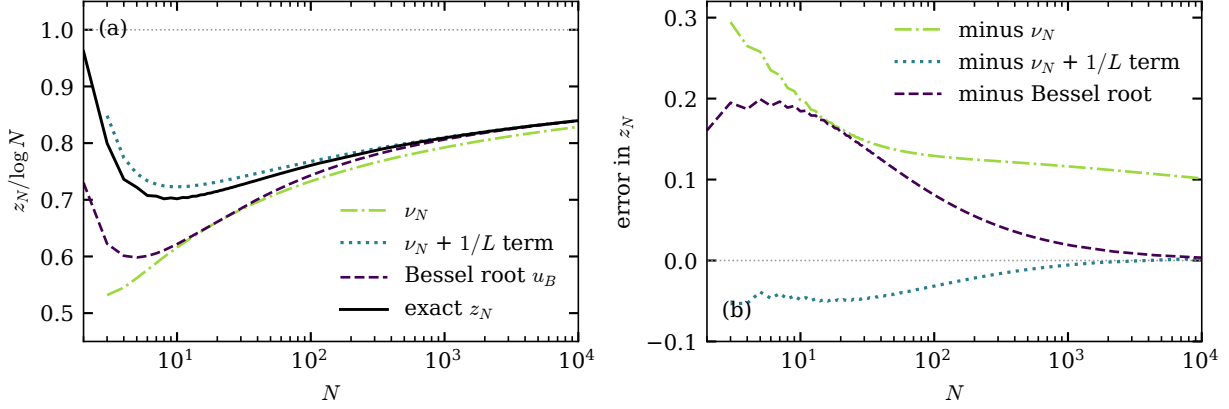


Figure 3: (a) The scaled crossing $z_N / \log N$ (solid) and the same quantity from the Bessel root u_B of Theorem 3.3 (dashed), from ν_N of (10) (through the constant) and from $\nu_N + (\frac{1}{4} \log L - \frac{1}{2} c_0 + \frac{1}{8}) / L$ of (11) (with the $1/L$ term). The ratio tends to 1 very slowly. (b) The exact z_N minus each approximation.

4 The crossing of every bin

The same question makes sense for each bin: at which p is bin k exactly as likely as an endpoint? Figure 4 shows the answer for 3 values of N . The crossing $1 - p_{a,N-a}$ decreases as the bin moves in from the edge, from order $N^{-1/2}$ next to the edge to order $(\log N)/N$ in the middle. This section proves this picture.

4.1 Ordering the crossings

Let $a, b \geq 1$ be integers, $N = a + b$ and $p > 0$, and recall that $S_{a,b}(u) = P_N(a)/P_N(0)$ with $u = (1 - p)/p$. Theorem 2.1 and $\binom{m-1}{r+1} = \frac{m-1-r}{r+1} \binom{m-1}{r}$ for $m = a, b$ give

$$S_{a,b}(u) = \sum_{r \geq 0} \binom{a-1}{r} \binom{b-1}{r} \left(2u^{2r+1} + \frac{N-2-2r}{r+1} u^{2r+2} \right). \quad (13)$$

Its coefficients are nonnegative, as $N - 2 - 2r \geq |b - a|$ for $r < \min(a, b)$, and the linear one is 2. So there is a unique $u_{a,b} > 0$ with $S_{a,b}(u_{a,b}) = 1$, and $p_{a,b} = 1/(1 + u_{a,b})$. As $S_{a,b}(u) \geq 2u$, we have $u_{a,b} \leq \frac{1}{2}$ and $p_{a,b} \in [\frac{2}{3}, 1)$, with $p_{a,b} = \frac{2}{3}$ only for $a = b = 1$.

Proposition 4.1 (ordering the crossings). *If $a < b - 1$, then*

$$u_{a+1,b-1} < u_{a,b}, \quad p_{a+1,b-1} > p_{a,b}.$$

Also $p_{a,b+1} > p_{a,b}$ for all positive a, b . In fact the corresponding inequalities between the polynomials $S_{a+1,b-1}$ and $S_{a,b}$, or $S_{a,b+1}$ and $S_{a,b}$, hold coefficientwise, with at least one strict coefficient.

In words: a bin closer to the middle, or in a longer row, needs a larger p to fall to the endpoint height.

Proof. For fixed N and $r \leq a-1$, $\binom{a-1}{r} \binom{b-1}{r} = (r!)^{-2} \prod_{j=0}^{r-1} (a-1-j)(b-1-j)$, a product of positive factors. Each factor grows when (a, b) becomes $(a+1, b-1)$, by $(a-j)(b-2-j) - (a-1-j)(b-1-j) = b - a - 1 > 0$. The even multiplier $(N - 2 - 2r)/(r + 1)$ is unchanged, and it is nonnegative where

either product is nonzero, since then $r \leq a$ and $N - 2 - 2r \geq b - a - 2 \geq 0$. New summands are nonnegative, and the cubic coefficient grows strictly, from $2(a-1)(b-1)$ to $2a(b-2)$. Increasing b with a fixed increases the binomial coefficients of Theorem 2.1, and the quadratic coefficient $N - 2$ strictly. A polynomial with nonnegative coefficients that dominates another coefficientwise, strictly somewhere, is strictly larger at every $u > 0$, so its root is smaller. $\square$

The ordering of the bin heights below also follows from the interior log-concavity of Dekking and Kong [12, Proposition 4.3]. We use the coefficients because they also give the strict gap in the next proposition.

Proposition 4.2 (adjacent bins and global extrema). *For $N \geq 2$ and $0 < p < 1$, an adjacent bin has the same probability as an endpoint precisely at*

$$p_N^{\text{adj}} = \frac{1 + \sqrt{N-1}}{2 + \sqrt{N-1}}.$$

For $0 < p < 1$, the endpoints are the least likely bins if $p < p_N^{\text{adj}}$, bins 1 and $N-1$ are if $p > p_N^{\text{adj}}$, and they tie at equality. For $N \geq 4$, the central crossing satisfies $p_N > p_N^{\text{adj}}$. At $p = p_N$, the middle and endpoints attain the global maximum and bins 1, $N-1$ attain the global minimum. For $N = 2, 3$, $p_N = p_N^{\text{adj}}$ and every bin ties at the crossing.

So at the central crossing the row has the shape of a W: high at the 2 ends and in the middle, and lowest next to the ends.

Proof. By (13), $S_{1,N-1}(u) = 2u + (N-2)u^2$, with positive root $u^{\text{adj}} = 1/(1 + \sqrt{N-1})$. This gives p_N^{adj} and the sign change. For $0 < p < 1$ we have $P_N(k) = P_N(0)S_{k,N-k}(u)$ with $P_N(0) > 0$, so if $1 \leq k < N-k-1$ Proposition 4.1 gives $P_N(k) < P_N(k+1)$. By symmetry the middle is the largest interior bin and bins 1, $N-1$ are the smallest, which gives the claims about the least likely bins. For $N \geq 4$ the cubic coefficient $2(\lfloor N/2 \rfloor - 1)(\lceil N/2 \rceil - 1)$ of S_N is positive, so $2u_N + (N-2)u_N^2 < S_N(u_N) = 1$, hence $u_N < u^{\text{adj}}$ and $p_N > p_N^{\text{adj}}$. At p_N the middle ties with the endpoints, which gives the global maximum, and bins 1, $N-1$ are below the endpoints, which gives the global minimum. For $N = 2, 3$ we have $S_N = S_{1,N-1}$, and symmetry gives the ties. $\square$

So $1 - p_N^{\text{adj}} \sim N^{-1/2}$, much larger than $1 - p_N \sim (\log N)/N$. The reason is simple. Away from the ends the single R of bin 1 needs 2 switches, so $S_{1,N-1} \approx Nu^2$ reaches 1 only at $u \approx N^{-1/2}$. (At $p = 0$ and $p = 1$ there are also ties between bins of probability 0.)

4.2 A Bessel estimate uniform in the bin

Bin a has about $(a^r/r!)(b^r/r!)$ words with r runs of each letter, each of weight about u^{2r} . The sum depends only on abu^2 , so a bin near the edge acts like the middle of a shorter row with $\sqrt{ab}$ in place of $N/2$. Put

$$s = \sqrt{ab}, \quad A = \frac{ab}{a+b}, \quad c = \frac{a+b}{2\sqrt{ab}}, \quad \eta = \frac{1}{c}, \quad x = su, \quad J_c(x) = I_0(2x) + cI_1(2x),$$

and $\hat{S}_{a,b}(u) = 2uJ_c(su)$. Here $c \geq 1$ and $0 < \eta \leq 1$, with $\eta = 1$ exactly when $a = b$.

Theorem 4.3 (uniform estimate for every bin). *For all positive integers a, b and all $u > 0$,*

$$0 \leq 1 - \frac{S_{a,b}(u)}{\widehat{S}_{a,b}(u)} \leq \frac{(a+b)u^2}{2} + u = \frac{x^2}{2A} + \frac{x}{s}. \quad (14)$$

In particular, for fixed N and u the bound does not depend on the bin. The inequality still holds when the right side exceeds 1, but then it gives no lower bound on $S_{a,b}$.

In words, every bin is a Bessel function of $\sqrt{ab}u$ up to a relative error of at most $Nu^2/2 + u$, which is $O(L^2/N)$ for every bin at once when u is about L/N . The proof is that of Theorem 3.3(i), with the 2 letters weighted by a and b .

Proof. Let $U_r = U_r(a)$ and $V_r = U_r(b)$ as in Lemma 3.2(b), and $H = 1/a + 1/b = 1/A$. Lemma 3.2(a), applied to the $2r$ clipped factors of $U_r V_r$, gives

$$0 \leq 1 - U_r V_r \leq \frac{H}{2} r(r+1). \quad (15)$$

The same argument for the weighted average $E_r = (aU_{r+1}V_r + bU_r V_{r+1})/(a+b)$ gives

$$\begin{aligned} 0 \leq 1 - E_r &\leq \frac{(r+1)(r+2)}{a+b} + \frac{r(r+1)}{2(a+b)} \left(\frac{a}{b} + \frac{b}{a} \right) \\ &= \frac{H}{2} r(r+1) + \frac{2(r+1)}{a+b}. \end{aligned} \quad (16)$$

Since $\binom{a-1}{r} = a^r U_r / r!$, (13) becomes

$$S_{a,b}(u) = 2u \left[\sum_{r \geq 0} \frac{x^{2r}}{(r!)^2} U_r V_r + c \sum_{r \geq 0} \frac{x^{2r+1}}{r!(r+1)!} E_r \right], \quad (17)$$

and dropping $U_r V_r$ and E_r gives $\widehat{S}_{a,b}(u)$. So by Lemma 3.2(c), with $I_j = I_j(2x)$, $(\widehat{S}_{a,b} - S_{a,b})/(2u)$ lies between 0 and

$$\frac{H}{2} (x^2 I_0 + x I_1 + c x^2 I_1) + \frac{2c x I_0}{a+b} = \left(\frac{x^2}{2A} + \frac{x}{s} \right) J_c(x),$$

since $H/2 = c/s = 1/(2A)$ and $2c/(a+b) = 1/s$. Divide by $\widehat{S}_{a,b}/(2u) = J_c(x)$. $\square$

To turn this into a bound on the crossing, let $x_{a,b} = s u_{a,b}$ and let x_B be the positive root of

$$G_\eta(x_B) = A, \quad G_\eta(x) = x \{I_1(2x) + \eta I_0(2x)\}. \quad (18)$$

Differentiating term by term gives

$$G'_\eta(x) - 2G_\eta(x) = 2x(1-\eta)\{I_0(2x) - I_1(2x)\} + \eta I_0(2x) \geq 0, \quad (19)$$

since $I_0(2x) - I_1(2x) = \frac{1}{\pi} \int_0^\pi e^{2x \cos \theta} (1 - \cos \theta) d\theta > 0$ by [28, Eq. 10.32.3]. So $(\log G_\eta)' \geq 2$ for $x > 0$ and $0 \leq \eta \leq 1$, and x_B is unique.

Corollary 4.4 (quantitative crossing bound). *We have $x_{a,b} \geq x_B$. If*

$$\delta_{a,b} = \frac{x_{a,b}^2}{2A} + \frac{x_{a,b}}{s} < 1,$$

then

$$0 \leq x_{a,b} - x_B \leq -\frac{1}{2} \log(1 - \delta_{a,b}). \quad (20)$$

Proof. Since $c\eta = 1$ and $2c/s = 1/A$, we have $\widehat{S}_{a,b}(u) = G_\eta(su)/A$. At the crossing $S_{a,b} = 1$, so Theorem 4.3 gives $A \leq G_\eta(x_{a,b}) \leq A/(1 - \delta_{a,b})$. As $G_\eta(x_B) = A$, this gives $x_{a,b} \geq x_B$, and integrating $(\log G_\eta)' \geq 2$ from x_B to $x_{a,b}$ gives (20). This is the argument of Theorem 3.3(ii). For even N and $a = b$ the 2 statements coincide, since then $\eta = 1$, $s = h$ and $2G_1 = F$. $\square$

Remark 4.5 (Jacobi polynomials). The Bessel forms in this paper are classical in another guise: they come from the confluence of Jacobi, Laguerre and Bessel functions near the point 1. For $a \leq b$ and $0 \leq t < 1$,

$$\sum_{r \geq 0} \binom{a-1}{r} \binom{b-1}{r} t^r = {}_2F_1(1-a, 1-b; 1; t) = (1-t)^{a-1} P_{a-1}^{(0, b-a)} \left(\frac{1+t}{1-t} \right),$$

and for $b > a$ and $0 \leq u < 1$,

$$\sum_{r \geq 0} \binom{a-1}{r} \binom{b-1}{r+1} u^{2r+2} = \frac{(b-1)u^2}{a} (1-u^2)^{a-1} P_{a-1}^{(1, b-a-1)} \left(\frac{1+u^2}{1-u^2} \right).$$

Near 1, Jacobi polynomials of large degree behave like Bessel functions (Mehler–Heine [28, Eq. 18.11.5]), with expansions in modified Bessel functions uniform near 1 by Elliott [13]. For fixed a and $y = bu^2$, as $b \rightarrow \infty$, [28, Eq. 18.7.21] turns the last sum into $(y/a)L_{a-1}^{(1)}(-y) = H_a(y)$ of (25), and the other terms tend to 0; see also Temme [27]. But Elliott’s expansions have fixed parameters and no error bounds, while here $b-a$ can be as large as $N-2$, and Theorem 4.3 gives an explicit relative error at every N .

4.3 Every growing distance from an edge

The Bessel root satisfies an equation of the form $z + \frac{1}{2} \log z = \ell + C + B/z + \dots$. The next lemma inverts such equations.

Lemma 4.6 (inverting $z + \frac{1}{2} \log z$). *Fix $K > 0$. There is a constant C_K with the following property. Let $z \geq 1$ and $\ell \geq 2$, and suppose that $z + \frac{1}{2} \log z = \ell + C + B/z + \omega$ with $|C|, |B| \leq K$ and $|\omega| \leq 1$. Then*

$$\left| z - \ell + \frac{1}{2} \log \ell - C - \frac{\frac{1}{4} \log \ell - \frac{1}{2} C + B}{\ell} \right| \leq C_K \left(\frac{(\log \ell)^2}{\ell^2} + |\omega| \right).$$

Proof. Put $w = z - \ell$. As $z \geq 1$, $\log z \geq 0$ and $|B/z| \leq K$, so $z \leq \ell + 2K + 1$ and $|w| \leq 2K + 1 + \frac{1}{2} \log(\ell + 2K + 1) = O_K(\log \ell)$. This proves the claim for ℓ bounded in terms of K , since then both sides are $O_K(1)$ and $(\log \ell)^2/\ell^2$ is bounded below. For larger ℓ we have $|w| \leq \ell/2$, so $\log z = \log \ell + w/\ell + O((w/\ell)^2)$ and $B/z = B/\ell + O_K(|w|/\ell^2)$, and $w = C - \frac{1}{2} \log \ell - w/(2\ell) + B/\ell + \omega + O_K((\log \ell)^2/\ell^2)$. This gives first $w = d + O_K(\log \ell/\ell + |\omega|)$ with $d = C - \frac{1}{2} \log \ell$, and then, putting this back into the term $w/(2\ell)$, $w = d + (B - d/2)/\ell + O_K((\log \ell)^2/\ell^2 + |\omega|)$. $\square$

Theorem 4.7 (uniform crossing expansion). *Let $1 \leq a \leq b$, and put*

$$\ell = \log A, \quad C(\eta) = \frac{1}{2} \log(8\pi) - \log(1 + \eta), \quad B(\eta) = \frac{3 - \eta}{8(1 + \eta)}.$$

As $a \rightarrow \infty$, uniformly over $b \geq a$,

$$\begin{aligned} 2\sqrt{ab} u_{a,b} &= \ell - \frac{1}{2} \log \ell + C(\eta) \\ &\quad + \frac{\frac{1}{4} \log \ell - \frac{1}{2} C(\eta) + B(\eta)}{\ell} + O \left(\frac{(\log \ell)^2}{\ell^2} + \frac{(\log a)^2}{a} \right). \end{aligned} \tag{21}$$

That is, there are a_0 and C such that the error term is at most C times the bracket for all $a \geq a_0$ and all $b \geq a$. The same holds with $1 - p_{a,b}$ in place of $u_{a,b}$.

In words: a bin far from both edges crosses where $2\sqrt{ab}u$ is about $\log A$, with $A = ab/(a+b)$ half the harmonic mean of a and b . There is no condition on how a compares with N , so this covers a bin at distance $\sqrt{N}$ from the edge as well as the middle. The idea of the proof: Corollary 4.4 moves the problem to the Bessel root x_B , and the large-argument form of I_0 and I_1 turns (18) into an equation that Lemma 4.6 solves.

Proof. Here $a/2 \leq A \leq a$, $s \geq a$ and $0 < \eta \leq 1$.

Step 1: the crossing is at $x_{a,b} \leq \log a$. At $x = \log a$ the bound of Theorem 4.3 is at most $2(\log a)^2/a$, while $G_\eta(x)/A \geq xI_1(2x)/a \rightarrow \infty$, both uniformly in b . So $S_{a,b} > 1$ there, and $x_{a,b} \leq \log a$ for all $a \geq a_0$ and all $b \geq a$. Then $\delta_{a,b} \leq 2(\log a)^2/a$, and Corollary 4.4 gives

$$x_{a,b} - x_B = O((\log a)^2/a). \quad (22)$$

Step 2: the Bessel root. Write $z = 2x_B$, so $z \rightarrow \infty$ as $A \rightarrow \infty$, uniformly in η . By the large-argument form of I_0 and I_1 [28, Eq. 10.40.1], $I_1(z) + \eta I_0(z) = (1+\eta)e^z(2\pi z)^{-1/2}(1 - B(\eta)/z + O(z^{-2}))$ uniformly in $0 \leq \eta \leq 1$. Taking logarithms in (18) gives $z + \frac{1}{2} \log z = \ell + C(\eta) + B(\eta)/z + O(z^{-2})$. As $|C(\eta)| \leq 2$ and $0 < B(\eta) \leq \frac{3}{8}$, Lemma 4.6 with $K = 2$ expands z .

Step 3: back to the crossing. By (22) the expansion moves to $2x_{a,b} = 2\sqrt{ab}u_{a,b}$. Finally $2s(u - u/(1+u)) \leq 2x^2/s = O((\log a)^2/a)$ at $u = u_{a,b}$, which gives the same expansion for $1 - p_{a,b} = u_{a,b}/(1 + u_{a,b})$. $\square$

Hence, uniformly in b ,

$$1 - p_{a,b} \sim \frac{\log a}{2\sqrt{ab}} \quad (a \rightarrow \infty, b \geq a). \quad (23)$$

If $a = o(b)$, then $\eta \rightarrow 0$ and $\log A = \log a + o(1)$, so

$$2\sqrt{ab}(1 - p_{a,b}) = \log a - \frac{1}{2} \log \log a + \frac{1}{2} \log(8\pi) + o(1). \quad (24)$$

If $a/N \rightarrow \alpha \in (0, \frac{1}{2}]$ at any rate, then

$$2\sqrt{a(N-a)}(1 - p_{a,N-a}) = \log N - \frac{1}{2} \log \log N + \log\{\alpha(1-\alpha)\} + C(2\sqrt{\alpha(1-\alpha)}) + o(1).$$

At the center, $a = b = N/2$ gives $A = N/4$ and $\eta = 1$, and changing from $\log A$ to $\log N$ recovers (11).

4.4 Fixed bins and the Laguerre edge law

Now let a be fixed and $b \rightarrow \infty$. A run of R 's away from the ends needs 2 switches, which gives a factor u^2 , and it can go in about b places. So the right variable is $y = bu^2$. Define

$$H_a(y) = \sum_{r=0}^{a-1} \binom{a-1}{r} \frac{y^{r+1}}{(r+1)!} = \frac{y}{a} L_{a-1}^{(1)}(-y), \quad (25)$$

with the standard Laguerre polynomial $L_{a-1}^{(1)}$. Let y_a be the positive root of $H_a(y_a) = 1$ and $c_a = \sqrt{y_a}$. Then $y_1 = 1$, $y_2 = \sqrt{3} - 1$, and y_a decreases strictly in a , by coefficient comparison.

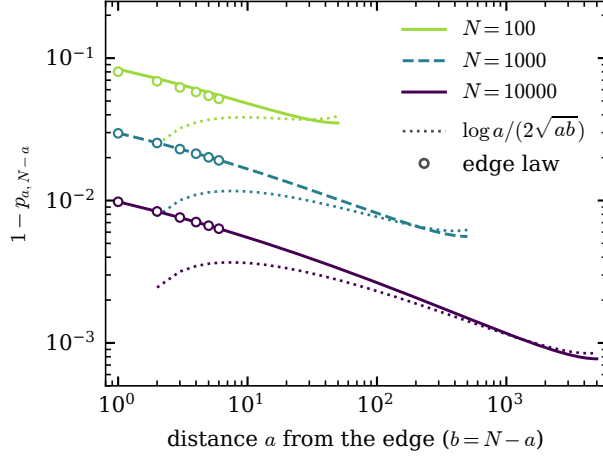


Figure 4: The crossing $1 - p_{a, N-a}$ of each bin against its distance a from the nearer edge, for $N = 100, 1000$ and 10^4 . The dotted lines are the leading law (23), and the circles are the edge law (27) for $a \leq 6$. The right end of each line is the central crossing $1 - p_N$.

Theorem 4.8 (fixed-edge expansion). *For each fixed positive integer a , as $b \rightarrow \infty$,*

$$u_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1}{b} + \frac{\gamma_a}{b^{3/2}} + O_a(b^{-2}), \quad (26)$$

where

$$\gamma_a = \frac{1 + y_a + (y_a + y_a^2/2)H_a''(y_a)/H_a'(y_a)}{2c_a}.$$

Thus the second coefficient in the odds expansion is -1 for every fixed bin. For the switching probability,

$$1 - p_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1 + c_a^2}{b} + O_a(b^{-3/2}). \quad (27)$$

In words, bin a near the edge crosses at odds about $\sqrt{y_a/b}$, and the first correction does not depend on a . For bin 1 this gives $u_{1,b} = b^{-1/2} - b^{-1} + b^{-3/2} + \dots$, the expansion of the exact root $1/(1 + \sqrt{b})$ of Proposition 4.2.

Proof. Put $\epsilon = b^{-1/2}$, $y = bu^2$ and $V_r = U_r(b)$. For $b \geq a$, (13) reads

$$\begin{aligned} S_{a,b}(u) &= 2\sqrt{y/b} \sum_r \binom{a-1}{r} \frac{y^r}{r!} V_r \\ &\quad + \frac{y}{b} \sum_r \binom{a-1}{r+1} \frac{y^r}{r!} V_r + y \sum_r \binom{a-1}{r} \frac{y^r}{(r+1)!} V_{r+1}. \end{aligned} \quad (28)$$

These sums have at most a terms, and with $V_r = 1$ they are $H_a'(y)$, $H_a''(y)$ and $H_a(y)$. Expanding $V_r = 1 - r(r+1)/(2b) + O_a(b^{-2})$ and using $\sum_r (r+1)(r+2) \binom{a-1}{r} \frac{y^{r+1}}{(r+1)!} = y^2 H_a''(y) + 2y H_a'(y)$ gives, uniformly near $y = y_a$,

$$\begin{aligned} S_{a,b}(u) &= H_a(y) + 2\sqrt{y} H_a'(y) \epsilon \\ &\quad + \{(y - y^2/2) H_a''(y) - y H_a'(y)\} \epsilon^2 + O_a(\epsilon^3). \end{aligned} \quad (29)$$

With $b = \epsilon^{-2}$ the right side of (28) is analytic in (ϵ, y) near $(0, y_a)$, and $H'_a(y_a) > 0$. So the implicit-function theorem gives an analytic solution $y(\epsilon) = y_a + d_1\epsilon + d_2\epsilon^2 + O_a(\epsilon^3)$ of $S_{a,b} = 1$, and by the uniqueness of the crossing it is the value of y at $u = u_{a,b}$. Matching coefficients in (29) gives $d_1 = -2c_a$ and $d_2 = 2 + y_a + (y_a + y_a^2/2)H''_a(y_a)/H'_a(y_a)$. Expanding $u = \epsilon\sqrt{y}$ and then $u/(1+u)$ proves both expansions. $\square$

For large a the Laguerre root y_a should obey the growing-distance law (24). The next theorem shows this.

Theorem 4.9 (matching the edge and Bessel regimes). *For every positive integer a and every $x > 0$,*

$$0 \leq 1 - \frac{aH_a(x^2/a)}{xI_1(2x)} \leq \frac{x^2}{2a}. \quad (30)$$

Consequently, with $\ell = \log a$ and $C_0 = \frac{1}{2} \log(8\pi)$, as $a \rightarrow \infty$,

$$\begin{aligned} 2\sqrt{ay_a} &= \ell - \frac{1}{2} \log \ell + C_0 \\ &\quad + \frac{\frac{1}{4} \log \ell - \frac{1}{2} C_0 + 3/8}{\ell} + O\left(\frac{(\log \ell)^2}{\ell^2} + \frac{(\log a)^2}{a}\right). \end{aligned} \quad (31)$$

Proof. With $U_r = U_r(a)$, $aH_a(x^2/a) = x \sum_{r \geq 0} U_r x^{2r+1}/(r!(r+1)!)$, so Lemma 3.2(b) and the second identity of Lemma 3.2(c) give (30). Put $x_a = \sqrt{ay_a}$, so $aH_a(x_a^2/a) = a$, and $x_a < \log a$ for large a by (30). So $a \leq G_0(x_a) \leq a/(1 - x_a^2/(2a))$, and as in Corollary 4.4, $x_a - x_0 = O((\log a)^2/a)$, where $G_0(x_0) = a$. So $z = 2x_0$ satisfies $z + \frac{1}{2} \log z = \ell + C_0 + 3/(8z) + O(z^{-2})$ by [28, Eq. 10.40.1], as in the proof of Theorem 4.7 with $\eta = 0$ and $A = a$, and Lemma 4.6 gives the expansion. $\square$

Figure 4 compares these laws with the exact crossings.

4.5 The moving boundary layer

The crossings also give the shape of a whole row at odds $u = v_N > 0$. Between the central crossing (v_N about $(\log N)/N$) and the adjacent one (v_N about $N^{-1/2}$), the middle is above the endpoints and a layer of bins next to each edge is below them. How thick is this layer? Let

$$\mathcal{L}_N = \#\{1 \leq k \leq \lfloor N/2 \rfloor : P_N(k) < P_N(0)\}.$$

Corollary 4.10 (boundary-layer width). *Suppose that, as $N \rightarrow \infty$,*

$$\Omega_N = \frac{1}{\sqrt{N} v_N} \rightarrow \infty, \quad \frac{\log \Omega_N}{N v_N} \rightarrow 0.$$

Then

$$\mathcal{L}_N \sim \Omega_N^2 \log^2 \Omega_N = \frac{\log^2(1/(\sqrt{N} v_N))}{N v_N^2}. \quad (32)$$

In particular, if $v_N = N^{-\gamma}$ with $1/2 < \gamma < 1$, then

$$\mathcal{L}_N \sim (\gamma - \frac{1}{2})^2 N^{2\gamma-1} \log^2 N.$$

By symmetry, the total number of interior bins below the endpoint height is $2\mathcal{L}_N$ for all sufficiently large N .

Proof. Set $d_N = \Omega_N^2 \log^2 \Omega_N$. Then $d_N \rightarrow \infty$ and $d_N/N = (\log \Omega_N / (N v_N))^2 \rightarrow 0$. Fix $\varepsilon \in (0, 1)$ and integers $a_{\pm} = (1 \pm \varepsilon)d_N + O(1)$. Then $\log a_{\pm} \sim 2 \log \Omega_N$ and $\sqrt{a_{\pm}(N - a_{\pm})} v_N \sim \sqrt{1 \pm \varepsilon} \log \Omega_N$, so (23) gives $u_{a_{\pm}, N - a_{\pm}}/v_N \rightarrow 1/\sqrt{1 \pm \varepsilon}$. Bin k is below the endpoint exactly when $v_N < u_{k, N-k}$, and $u_{k, N-k}$ decreases strictly for $k \leq N/2$ by Proposition 4.1. So $a_- \leq \mathcal{L}_N < a_+$ for large N , and we let $\varepsilon \downarrow 0$. Finally $\mathcal{L}_N = o(N)$, so the reflected intervals are disjoint. $\square$

5 Periodic boundaries and the crossing window

There are 2 ways to make Kagey's board periodic, and they are different. On a cylinder the ball moves around a circle of pegs, and its position becomes uniform (Remark 2.7). Here we keep the line of bins but close the sequence of steps into a loop, so that the last step also interacts with the first. This is the periodic Ising chain, with weight proportional to $\exp(J \sum_{i=1}^N \sigma_i \sigma_{i+1})$, $\sigma_{N+1} = \sigma_1$ and again $u = e^{-2J}$. Antal, Droz and Rácz [5] give its magnetization law at a fixed scale, and Dantchev and Rudnick [11] give exact coefficients. We ask how much the closing bond moves the central crossing, and then look at the row near the crossing under both boundaries.

Write $R_{N,k}^F = S_{k, N-k}$ and $R_{N,k}^P$ for the bin-to-endpoint ratios under free and periodic boundaries. Put $a = k$, $b = N - k$ and $s = \sqrt{ab}$. The classical cyclic count is

$$R_{N,k}^P(u) = \sum_{r=1}^{\min(a,b)} \frac{N}{r} \binom{a-1}{r-1} \binom{b-1}{r-1} u^{2r}. \quad (33)$$

To see this, note that a cyclic word with a letters R and b letters L switches an even number $2r$ of times and has r runs of each letter. Choose compositions of a and of b into r positive runs and a position for the start of a distinguished run of R's, and divide by the r choices of that run. The normalization of the periodic measure cancels in the ratio. In particular $R_{N,1}^P = Nu^2$, which gives the adjacent crossing $J = (\log N)/4$ of Stepanyan et al. [26]. Every interior ratio increases strictly from 0 to ∞ in $u > 0$, and the ratios increase strictly toward the middle, apart from the symmetric middle tie, by the coefficient comparison of Proposition 4.1.

Theorem 5.1 (periodic relative error). *For $N \geq 2$, $1 \leq k \leq N - 1$, and $u > 0$, define*

$$\widehat{R}_{N,k}^P(u) = \frac{Nu}{\sqrt{k(N-k)}} I_1\left(2u\sqrt{k(N-k)}\right).$$

Then

$$0 \leq 1 - \frac{R_{N,k}^P(u)}{\widehat{R}_{N,k}^P(u)} \leq \frac{Nu^2}{2}. \quad (34)$$

In words, under periodic boundaries every bin is a single Bessel function I_1 , with the same kind of error as in Theorem 4.3.

Proof. Put $x = su$, $A = ab/N$, $U_r = U_r(a)$ and $V_r = U_r(b)$. Shifting the index in (33) gives $R_{N,k}^P(u) = (Nu/s) \sum_{r \geq 0} U_r V_r x^{2r+1} / (r!(r+1)!)$, so the ratio to $\widehat{R}_{N,k}^P$ is an average of the $U_r V_r$ with weights $x^{2r+1} / (r!(r+1)!)$, whose sum is $I_1(2x)$. By (15), $0 \leq 1 - U_r V_r \leq r(r+1)/(2A)$, and the second identity of Lemma 3.2(c) bounds the deficit by $x^2/(2A) = Nu^2/2$. $\square$

We can guess the next theorem. A cyclic word switches an even number of times, so only I_1 appears in the periodic approximation, while the free words that start and end on different letters

give the extra I_0 term of the free one. As $I_0 \approx I_1$ for large arguments, the free ratio is about twice the periodic one near the crossing. Both grow like e^{Nu} , so the periodic crossing needs Nu about $\log 2$ larger. The same factor 2 appears at a fixed scale T in the endpoint atoms $1/(2 \cosh T)$ (periodic) and $e^{-T}/2$ (free) of Antal, Droz and Rácz [5, Eqs. (37) and (47)], while their densities at zero magnetization agree to leading order.

Theorem 5.2 (periodic crossing and boundary shift). *Let u_N^P be the positive root of $R_{N,m_N}^P(u) = 1$, let $p_N^P = 1/(1 + u_N^P)$, and let $u_N^F = u_N$ be the free central root. Put $c_P = \frac{1}{2} \log(\pi/2)$. As $N \rightarrow \infty$,*

$$Nu_N^P = L - \frac{1}{2} \log L + c_P + \frac{\frac{1}{4} \log L - \frac{1}{2} c_P + \frac{3}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right), \quad (35)$$

$$N(u_N^P - u_N^F) = \log 2 + \frac{\frac{1}{4} - \frac{1}{2} \log 2}{\log N} + o(1/\log N). \quad (36)$$

Replacing Nu_N^P by $N(1 - p_N^P)$ preserves (35). At the periodic crossing the coupling $J_N^P = -\frac{1}{2} \log u_N^P$ satisfies

$$J_N^P = \frac{1}{2}(\log N - \log \log N) + \frac{\frac{1}{4} \log \log N - \frac{1}{2} c_P}{\log N} + O\left(\frac{(\log \log N)^2}{(\log N)^2}\right). \quad (37)$$

In words, closing the loop makes every nonconstant word switch an even number of times, so the endpoints become relatively more likely, and the periodic chain needs about $\log 2$ more expected switches before the middle catches up with them. Since $c_P = c_0 + \log 2$, the periodic expansion is the free one shifted by $\log 2$, with $\frac{3}{8}$ in place of $\frac{1}{8}$ in the $1/L$ term.

Proof. Step 1: the first order. At the middle $2s = N + O(1/N)$. At $u = (1 \pm \epsilon)L/N$ with fixed $\epsilon \in (0, 1)$ the bound of Theorem 5.1 tends to 0, and the large-argument form of I_1 makes $\hat{R}^P$ tend to ∞ with the plus sign and to 0 with the minus sign. Since R_{N,m_N}^P increases in u , we get $Nu_N^P \sim L$.

Step 2: the crossing equation. Write $z = Nu_N^P$. At the root $\log \hat{R}^P = O(L^2/N)$ by (34). Replacing the argument $2su$ by z changes it by $O(L/N^2)$, and $I_1(z) = e^z(2\pi z)^{-1/2}(1 - 3/(8z) + O(z^{-2}))$ [28, Eq. 10.40.1]. So

$$z + \frac{1}{2} \log z = L + c_P + \frac{3}{8z} + O(z^{-2} + L^2/N),$$

and Lemma 4.6 with $\ell = L$, $C = c_P$ and $B = \frac{3}{8}$ gives (35). As $N(u - u/(1 + u)) = O(L^2/N)$ for $Nu = O(L)$, it also holds for $N(1 - p_N^P)$.

Step 3: the shift and the coupling. By (2), $Nu_N = z_N + O(L^2/N)$. Subtracting (11), where $c_0 = c_P - \log 2$ and the correction is $\frac{1}{8}$, gives (36), even with error $O((\log L)^2/L^2)$. Finally $J = -\frac{1}{2} \log u = \frac{1}{2}(L - \log z)$, and $\log z = \log L - (\frac{1}{2} \log L - c_P)/L + O((\log L)^2/L^2)$ gives (37). $\square$

Stepanyan et al. [26, Eq. (16)] approximate the periodic crossing by $\beta J \approx (2 \log N - 1)/5$, and their Fig. 1(b) fits the line $\beta J \approx 0.3864 \log N - 0.21055$. At $N = 10^5$ the exact periodic crossing is $J = 4.576$ (their βJ). Their Eq. (16) gives 4.405, their line gives 4.238, and (37) without its error term gives 4.578. The line is accurate for $10 \leq N \leq 100$ (at $N = 100$ it gives 1.569 and the exact value is 1.574), but it has no $\log \log N$ term.

Now we look at the whole row near the crossing. How many bins are above the endpoints just past the crossing, and how does their height fall off away from the middle? The factor e^{-2y^2} in the next theorem is easy to guess. Moving $yN/\sqrt{\log N}$ away from the middle multiplies the Bessel argument by $2\sqrt{k(N-k)}/N \approx 1 - 2y^2/\log N$. The argument is about $\log N$, so it drops by about $2y^2$, and the ratio, which grows like e to the argument, drops by about the factor e^{-2y^2} .

Theorem 5.3 (spatial crossing window). *Let $X \in \{F, P\}$ and let u_N^X be its exact central crossing. For fixed real t, y set*

$$u = u_N^X + \frac{t}{N}, \quad k_N(y) = \text{nint}\left(\frac{N}{2} + \frac{yN}{\sqrt{\log N}}\right),$$

where either nearest-integer convention may be used. Then, as $N \rightarrow \infty$, uniformly on compact sets of (t, y) ,

$$R_{N, k_N(y)}^X(u_N^X + t/N) \rightarrow e^{t-2y^2}. \quad (38)$$

If $M_N^X(t)$ is the number of interior bins with probability strictly above the endpoint probability at that parameter, then

$$\frac{\sqrt{\log N}}{N} M_N^X(t) \rightarrow \sqrt{2 \max(t, 0)}. \quad (39)$$

In words, the crossing window has width about $1/N$ in u and about $N/\sqrt{\log N}$ in space, and inside it the row is a Gaussian bump times e^t . A weak limit theorem cannot give this, since it says nothing about a single bin such as the endpoint.

Proof. Step 1: the profile. Put $\delta = k/N - \frac{1}{2}$ and $\eta = 2\sqrt{k(N-k)}/N = \sqrt{1-4\delta^2}$. For the stated rounding and bounded y ,

$$\eta = 1 - \frac{2y^2}{L} + O\left(L^{-2} + \frac{1}{N\sqrt{L}}\right), \quad Nu_N^X = L + O(\log L).$$

So the Bessel argument $Nu\eta$ at (u, k) is $Nu_N^X + t - 2y^2 + o(1)$, uniformly on the compact set. By Theorems 4.3 and 5.1 both ratios are within relative error $O(L^2/N)$ of their Bessel forms, and $I_0(z)$ and $I_1(z)$ equal $e^z(2\pi z)^{-1/2}(1 + O(1/z))$ [28, Eq. 10.40.1], so the prefactors relative to their central values tend to 1 uniformly. The change of argument gives e^{t-2y^2} , and the exact central ratio is 1, which proves (38).

Step 2: the count. The interior ratios increase toward the middle. If $t \leq 0$, the central ratio is at most 1, so no interior bin is above the endpoint. Let $t > 0$ and $0 < \epsilon < \sqrt{t/2}$. By Step 1, for large N the bins at scaled distance less than $\sqrt{t/2} - \epsilon$ are above the endpoint and those at distance $\sqrt{t/2} + \epsilon$ are below, and by monotonicity so are all farther bins. Count the bins in between, divide by $N/\sqrt{L}$ and let $\epsilon \downarrow 0$. Rounding changes the count by $O(1)$. $\square$

Figure 5 shows the exact ratios at $N = 10^6$.

6 A uniform local law

So far we compared each bin with an endpoint. Now we approximate $P_N(k)$ itself by a single formula. This is a local form of Kagey's normal limit, but we want it uniform in $q \leq \frac{1}{2}$: in the diffusive regime (q fixed) the bin is close to Gaussian, while in the ballistic regime ($T = Nq$ bounded) the walk switches only a bounded number of times and the Gaussian form fails (Proposition 6.4). We first write $P_N(k)$ exactly through a difference of 2 binomial variables, then replace that difference by a Skellam law with the same mean and variance. Throughout, $n_1 = k - 1$, $n_2 = N - k - 1$ and $n_{\min} = \min(n_1, n_2)$.

Theorem 6.1 (a difference of 2 binomials). *Let $N \geq 2$, $1 \leq k \leq N - 1$ and $0 < q < 1$. Let $Y_1 \sim \text{Bin}(n_1, q)$ and $Y_2 \sim \text{Bin}(n_2, q)$ be independent, and put $D = Y_1 - Y_2$. Then*

$$P_N(k) = \frac{q}{2} [2\mathbb{P}(D = 0) + \mathbb{P}(D = 1) + \mathbb{P}(D = -1)]. \quad (40)$$

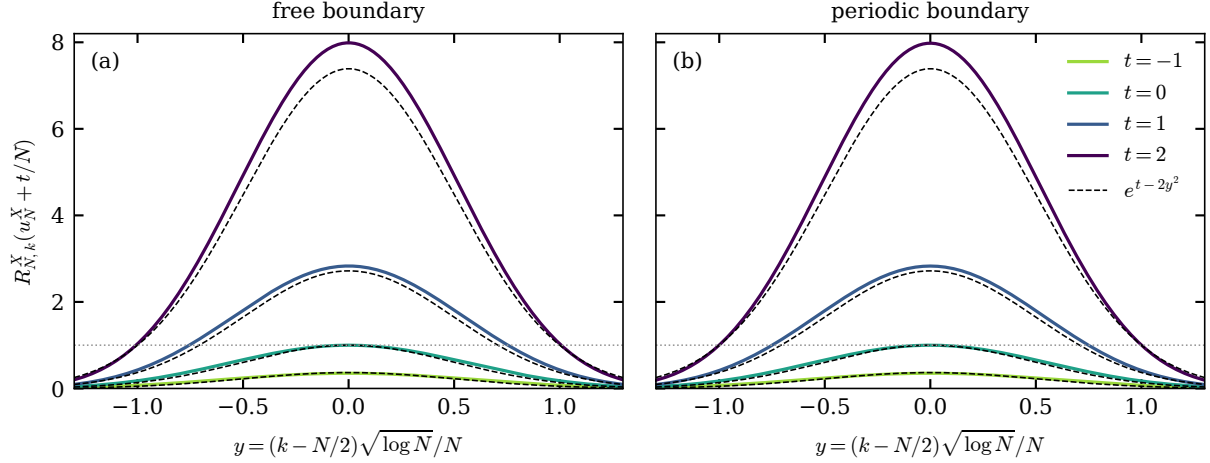


Figure 5: The exact ratios $R_{N,k}^X(u_N^X + t/N)$ at $N = 10^6$ for free (a) and periodic (b) boundaries, with the limit e^{t-2y^2} of Theorem 5.3 (dashed). Here $y = (k - N/2)\sqrt{\log N}/N$, and the bins above the dotted line are above the endpoint height. The convergence is slow, and at $t = 2$ and $y = 0$ the relative error is still 0.08.

In words, $Y_1 + 1$ and $Y_2 + 1$ play the roles of the numbers of runs of R's and of L's. In a word these differ by at most 1, which is why only $D \in \{-1, 0, 1\}$ appears, and the factor q pays for the 1 switch that every interior bin needs.

Proof. We rewrite Theorem 2.1, where $a - 1 = n_1$ and $b - 1 = n_2$. Note that $\mathbb{P}(Y_1 = r)\mathbb{P}(Y_2 = r') = \binom{n_1}{r}\binom{n_2}{r'}q^{r+r'}p^{N-2-r-r'}$. The term $j = 2r + 1$ of Theorem 2.1 is $\binom{n_1}{r}\binom{n_2}{r}p^{N-2-2r}q^{2r+1} = q\mathbb{P}(Y_1 = r)\mathbb{P}(Y_2 = r)$, and summing over r gives $q\mathbb{P}(D = 0)$. The term $j = 2r + 2$ is

$$\frac{1}{2} \left[\binom{n_1}{r+1}\binom{n_2}{r} + \binom{n_1}{r}\binom{n_2}{r+1} \right] p^{N-3-2r}q^{2r+2} = \frac{q}{2} [\mathbb{P}(Y_1 = r+1)\mathbb{P}(Y_2 = r) + \mathbb{P}(Y_1 = r)\mathbb{P}(Y_2 = r+1)],$$

and summing over r gives $\frac{q}{2}[\mathbb{P}(D = 1) + \mathbb{P}(D = -1)]$. $\square$

For $k = 1$ we have $n_1 = 0$ and $D = -Y_2$, so $\mathbb{P}(D = 0) = p^{N-2}$, $\mathbb{P}(D = -1) = (N-2)qp^{N-3}$ and $\mathbb{P}(D = 1) = 0$. By (40) and symmetry,

$$P_N(1) = P_N(N-1) = qp^{N-2} + \frac{1}{2}(N-2)q^2p^{N-3}. \quad (41)$$

With $P_N(0) = P_N(N) = \frac{1}{2}p^{N-1}$, the 4 bins $0, 1, N-1, N$ are exact, and from now on $N \geq 4$ and $2 \leq k \leq N-2$, so that $n_1, n_2 \geq 1$.

Next we tilt D to mean 0. For $s > 0$ put

$$r_1 = \frac{qs}{p+qs}, \quad r_2 = \frac{q}{ps+q}, \quad v_i = r_i(1-r_i), \quad \Lambda(s) = (p+qs)^{n_1}(p+q/s)^{n_2}.$$

Under the tilted law $\mathbb{P}_s$, $Y_1 \sim \text{Bin}(n_1, r_1)$ and $Y_2 \sim \text{Bin}(n_2, r_2)$ are independent, and $\mathbb{P}(D = j) = \Lambda(s)s^{-j}\mathbb{P}_s(D = j)$ (Lemma B.1). There is a unique $s > 0$ with $n_1r_1 = n_2r_2 =: m$, so that $\mathbb{E}_s D = 0$ (Lemma B.2). We fix this s , let $x = n_1v_1 + n_2v_2$ be the variance of D under $\mathbb{P}_s$, and define

$$U_N(k) = \frac{q}{2}\Lambda(s)e^{-x}[2I_0(x) + (s+s^{-1})I_1(x)]. \quad (42)$$

At the center of an even row, $n_1 = n_2$ gives $s = 1$, $r_1 = r_2 = q$ and $\Lambda = 1$, so

$$U_N(N/2) = q e^{-x} (I_0(x) + I_1(x)), \quad x = (N-2)pq. \quad (43)$$

Here is what (42) does. By (40) and the tilt, $P_N(k) = \frac{q}{2}\Lambda(s)[2\mathbb{P}_s(D=0) + s^{-1}\mathbb{P}_s(D=1) + s\mathbb{P}_s(D=-1)]$. The law $e^{-x}I_{|j|}(x)$ on $\mathbb{Z}$ is the symmetric Skellam law, the law of a difference of 2 independent Poisson variables with mean $x/2$, and it has mean 0 and variance x . So $U_N(k)$ replaces the tilted D by the Skellam law with the same mean and variance. This law is exact for a Poisson difference, which D resembles when q is small, and close to Gaussian when x is large. At the center D is a sum of $N/2 - 1$ independent variables with $\mathbb{P}(\pm 1) = pq$, and replacing it by this Skellam law is the approximation that Čekanavičius and Liaudanskaitė [10, Eq. (3)] attribute to Presman. The tilt extends it off the center.

To state the error bound, let $\kappa = 1 - \cos t$ and write $\langle h \rangle$ for the average of $h(t)$ against the density proportional to $e^{x \cos t}$ on $[-\pi, \pi]$. With $\mathcal{R} = \mathcal{R}(x) = I_1(x)/I_0(x)$, Lemma B.6 gives

$$\langle \kappa^2 \rangle = 2 - 2\mathcal{R} - \frac{\mathcal{R}}{x}, \quad \langle \kappa^3 \rangle = 4 - 4\mathcal{R} - \frac{3\mathcal{R}}{x} - \frac{2\mathcal{R}}{x^2} + \frac{1}{x}.$$

Put $V = n_1 v_1^2 + n_2 v_2^2$ and $M = m \max(r_1, r_2)$.

Theorem 6.2 (uniform local law). *Let $N \geq 4$, $2 \leq k \leq N-2$ and $0 < q < 1$. Then $P_N(k) = U_N(k)(1 + \theta)$ with*

$$|\theta| \leq \mathcal{E} := \frac{1}{\mathcal{R}(x)} \left[\left(4V + \frac{64}{3}M \right) \langle \kappa^2 \rangle + \frac{1024}{9}M^2 \langle \kappa^3 \rangle \right].$$

In words, the relative error is controlled by 2 things: how far the tilted binomial steps are from Poisson (V , which is small when the r_i are small) and how skewed they are (M), each weighted by a moment of κ that is small when the variance x is large.

Idea of the proof. By Fourier inversion, $P_N(k)$ and $U_N(k)$ are $\frac{q}{2}\Lambda(s)\frac{1}{2\pi}\int_{-\pi}^{\pi} f\omega dt$ with $\omega(t) = 2 + se^{it} + s^{-1}e^{-it}$, where f is $\chi(t) = \mathbb{E}_s e^{itD}$ for $P_N(k)$ and the Skellam characteristic function $g(t) = e^{x(\cos t - 1)}$ for $U_N(k)$. Since $|1 - r + re^{it}|^2 = 1 - 2v\kappa = e^{-2v\kappa}(1 + O(v^2\kappa^2))$ with $v = r(1-r)$, we get $|\chi| \leq g$ and $g - |\chi| = O(V\kappa^2)g$. The phase of χ has no linear term since $\mathbb{E}_s D = 0$, so it is $O(M|t|^3)$. Hence $|\theta|$ is at most a constant times $((V + M)\langle \kappa^2 \rangle + M^2\langle \kappa^3 \rangle)/\mathcal{R}(x)$, where $\langle \kappa^j \rangle = O(\min(1, x^{-j}))$ and $1/\mathcal{R}(x) \leq 1 + 2/x$. Appendix B gives the proof with explicit constants.

Corollary 6.3. *Let $N \geq 4$, $2 \leq k \leq N-2$ and $0 < q \leq \frac{1}{2}$, and put $r_{\max} = \max(r_1, r_2)$. Then*

$$|\theta| \leq \min \left\{ \frac{22334}{n_{\min}}, 221 r_{\max} + 5825 r_{\max}^2 \right\}.$$

Also $r_{\max} \leq \delta_T := 2T/\sqrt{N} + 4T^2/N$, so $|\theta| \leq 221\delta_T + 5825\delta_T^2$ for every bin $2 \leq k \leq N-2$. At the center of an even row, $r_{\max} = q$ and $n_{\min} = N/2 - 1$.

So the relative error tends to 0 uniformly in $0 < q \leq \frac{1}{2}$ as $n_{\min} \rightarrow \infty$, and uniformly over $2 \leq k \leq N-2$ as $T/\sqrt{N} \rightarrow 0$, which covers the ballistic regime $T = O(1)$ up to the edges. The constants are crude. The largest $n_{\min}|\theta|$ we observed is 0.686 (at $N = 10^4$, $q = \frac{1}{2}$, over $2 \leq k \leq 400$ with $q = \frac{1}{2}$ or 0.3, and over every bin of our grid with $N \leq 1000$ and $10^{-5} \leq q \leq \frac{1}{2}$), while the first bound has the constant 22334 and exceeds 1 unless $n_{\min} > 22334$. Near an edge the largest error we observed is $|\theta| = 0.1617$ ($k = 3$, $N = 10^4$, $q = \frac{1}{2}$), an observation and not a bound. For $q > \frac{1}{2}$ Theorem 6.2 still holds, but $\mathcal{E}$ need not be small, and a uniform form is open. Near $q = 1$

the symmetric kernel fails once $n_1 \neq n_2$. We found $\theta = 0.578$ at the middle bin $k = 50$ of the odd row $N = 101$ with $q = 0.999$, while $\theta \approx 0$ at $N = 100$, $k = 50$.

Why not a Gaussian form? Replacing $e^{-x}I_j(x)$ in (42) by $(2\pi x)^{-1/2}$ gives the Gaussian saddle point $G_N(k) = \frac{q}{2}\Lambda(s)(2 + s + s^{-1})(2\pi x)^{-1/2}$. We also define the transfer-matrix form $\text{TM}_N = 2(2\pi(N-1)p/q)^{-1/2}$, the Gaussian local form for $\mathbb{P}(X_N = 0)$ with the variance $(N-1)p/q$ that λ_+^{N-1} gives in the proof of Theorem 2.5. The factor 2 is the span of the lattice of $X_N = 2k - N$.

Proposition 6.4 (Gaussian saddle points). *Fix an even $N \geq 4$. As $q \rightarrow 0$,*

$$\frac{P_N(N/2)}{q} \rightarrow 1, \quad \frac{G_N(N/2)}{P_N(N/2)} \sim \frac{2}{\sqrt{2\pi(N-2)q}} \rightarrow \infty, \quad \frac{\text{TM}_N}{P_N(N/2)} \sim \frac{2}{\sqrt{2\pi(N-1)q}} \rightarrow \infty.$$

Proof. The term $j = 1$ of Theorem 2.1 is qp^{N-2} and the other terms are $O(q^2)$, so $P_N(N/2) = q + O(q^2)$. At the center $s = 1$, $\Lambda = 1$ and $x = (N-2)pq$, so $G_N(N/2) = 2q/\sqrt{2\pi(N-2)pq}$. Also $\text{TM}_N = 2q/\sqrt{2\pi(N-1)pq}$. Let $q \rightarrow 0$. $\square$

More generally, since $P_N(N/2)/q \rightarrow 1$, any approximation $A_N(q)$ of the center with $A_N(q)/q \rightarrow \infty$ has relative error $P_N(N/2)/A_N(q) - 1 \rightarrow -1$. Both saddle points are of this kind. We make no claim about other Gaussian forms. The Bessel form avoids this because $e^{-x}I_j(x)$ is exact for the Poisson difference that D approaches in the ballistic regime.

Remark 6.5 (the center crossing). For even $N \geq 4$ let $\mathcal{G}(q)$ be the logarithm of the ratio of $U_N(N/2)$ to $P_N(0) = \frac{1}{2}p^{N-1}$. By $I'_0 = I_1$, $I'_1 = I_0 - I_1/x$ [28, Eqs. 10.29.2, 10.29.3] and $I_1 \leq I_0$, we get $\mathcal{G}'(q) \geq \frac{1}{2pq} + \frac{N-1}{p} > N-1$ on $(0, \frac{1}{2}]$. Also $\mathcal{G} \rightarrow -\infty$ as $q \rightarrow 0$ and $\mathcal{G}(\frac{1}{2}) \geq (N-1)\log 2 - \frac{N-2}{4} > 0$, so $\mathcal{G}$ has a unique root q_U there, an approximate crossing. In Table 3, which is purely numerical, the relative error of q_U is 16 to 36 times smaller than that of the Bessel root u_B of Theorem 3.3. Put $\vartheta_N = 221q_N + 5825q_N^2$. If $\vartheta_N < 1$, Corollary 6.3 gives $|\mathcal{G}(q_N)| \leq \vartheta_N/(1 - \vartheta_N)$, and the mean value theorem gives $|q_U - q_N| \leq \vartheta_N/((1 - \vartheta_N)(N-1))$. By our computation $\vartheta_N \geq 1$ for every even $N < 1454$, and beyond it this bound is weaker than Theorem 3.3(ii) at every N we computed.

N	p_N	$q_U/q_N - 1$	$u_B/u_N - 1$	ratio
100	0.9649663263	-1.477×10^{-3}	-2.39×10^{-2}	0.062
400	0.9881246429	-3.083×10^{-4}	-7.61×10^{-3}	0.041
1600	0.9962346137	-6.469×10^{-5}	-2.31×10^{-3}	0.028

Table 3: The root q_U of $U_N(N/2) = P_N(0)$ against the Bessel root u_B of Theorem 3.3. The ratio is $(q_U/q_N - 1)/(u_B/u_N - 1)$. Numerical values, not bounds.

Related local limits. Čekanavičius and Liaudanskaitė [10] and Čekanavičius and Vellaisamy [9] give total-variation, local and Wasserstein bounds with unspecified absolute constants for chains outside our regime. One is a symmetric 3-state chain whose step is 0 with probability at least $\frac{14}{15}$. The other is a 2-state chain with $\mathbb{P}(1 | 1) \leq \frac{1}{2}$, and some of its bounds also need $\mathbb{P}(1 | 0) \leq \frac{1}{30}$. Neither allows both states to persist, as our $p \rightarrow 1$ does. Jensen [18] conditions the count S_n on $0 < S_n < n$ and patches small tilted variances, and his relative error stays bounded (about 0.18, found partly by numerics) but does not tend to 0. In the literature we searched we did not find a tilted Skellam kernel off the center with a relative local error, explicit constants and uniformity in $q \leq \frac{1}{2}$.

7 Verification

Apart from Remark 3.9, which is computer-assisted throughout, the proofs rest on only 2 computations, both described where they are used: the integer certificates for $N \leq 174$ in Theorem 3.4 (Step 4) and Proposition 3.8, and the evaluation at $N = 2774$ in Corollary 3.7(d). The checks for all 4 papers are described in `SUPPLEMENT.md`, in the Problem 131 folder of <https://github.com/apemm/Kagey-Problems>, together with the manuscript source and the recorded results. The scripts are standard-library Python, and `python -B verify_all.py --paper a` runs the ones for this paper (`--full` adds 2 long runs of about 21 minutes). They check the exact law, the lattice-path counts, the orderings, the Bessel bounds and the asymptotic formulas, and where possible each reported number is computed in 2 independent ways. Before the main computations (the monotonicity and Bessel-ratio checks, the every- N certificates and the local law) we wrote predictions and conditions that would refute them in `ledger.md`, `paperA/data/ledger_every_n.md` and `paperA/data/ledger_llt.md`, and we recorded the outcomes there. Some small- N predictions for Theorem 3.4 were not blind, as the ledger says.

The Lean 4 proofs are in `lean/PaperA`, and `lean/README.md` lists them. They include the exact law, the numerator tables, the variance and the normal limit, the orderings of the crossings, the uniform Bessel bounds and root comparisons (`UniformBessel.lean`, `RootBounds.lean`), the leading terms $1 - p_N \sim (\log N)/N$ and $2\sqrt{ab}u_{a,b} \sim \log a$, the finite parts of the edge and periodic results, Lemmas 3.5 and 3.6 and Theorem 3.4 for every $N \geq 2$ (`BesselIntegral.lean`, `EveryN.lean`, `EveryNLarge.lean`). The finite parts are the polynomial H_a and its root y_a , the exact series (28), the cyclic count (33) and the adjacent periodic crossing. For Theorem 4.8 only the algebra of the coefficients is in Lean. For $N \leq 174$ the Lean kernel checks the same certificates in exact integer arithmetic, with its own rational bounds on ζ_N , so the whole theorem is proved in Lean. `lean/build-log.txt` shows that every theorem uses only the standard axioms. Remarks 2.6, 2.7, 4.5 and 3.9, Corollary 3.7 beyond part (a), Proposition 3.8, Lemma 4.6, Theorem 4.7 beyond its leading term, the expansions of Theorem 4.8, the expansion (31) of Theorem 4.9, Theorem 5.2, even its leading term, Corollary 4.10, Theorem 5.3 and Section 6 are not in Lean.

Acknowledgment. The Lean code for this paper was generated by AI. AI was also used to revise some of the writing and to help cross-verify the code.

A The crossing at every N

This appendix proves Lemma 3.6, the analytic half of Theorem 3.4, the identity behind the certificates and Proposition 3.8. In it the letters $U, W, W', r, r', t, t', \tau, \kappa, \ell, m, B, D, \beta, \Delta$ and δ have the local meanings given below. Each decimal constant is rounded in the safe direction.

A.1 Proof of Lemma 3.6

For integer n , $I_n(z) = \frac{1}{\pi} \int_0^\pi e^{z \cos t} \cos(nt) dt$ [28, Eq. 10.32.3], so $I_0(z) + I_1(z) = \frac{1}{\pi} \int_0^\pi e^{z \cos t} (1 + \cos t) dt$. Substitute $s = 1 - \cos t$, so that $\sin t = \sqrt{s(2-s)}$ and $1 + \cos t = 2 - s$, and then $s = v/z$. This gives

$$E(z) = \frac{1}{\sqrt{\pi}} \int_0^{2z} e^{-v} v^{-1/2} \sqrt{1 - \frac{v}{2z}} dv. \quad (44)$$

On $[0, 1]$ we have $\sqrt{1-y} = 1 - \sum_{k \geq 1} c_k y^k$ with every $c_k > 0$, $c_1 = \frac{1}{2}$ and $c_2 = \frac{1}{8}$, and $\sum_k c_k = 1$ since the series tends to 0 at $y = 1$. Since $\sum_{k \geq 3} c_k y^k \leq \frac{3}{8} y^3$, for $0 \leq y \leq 1$

$$q(y) := 1 - \frac{y}{2} - \frac{y^2}{8} - \frac{3y^3}{8} \leq \sqrt{1-y} \leq 1 - \frac{y}{2} - \frac{y^2}{8} \leq p(y) := 1 - \frac{y}{2} - \frac{y^2}{8} + \frac{y^3}{8}.$$

The cubic q decreases and $q(1) = 0$, so $q \leq 0$ on $[1, \infty)$. Also $8p(y) = (y-2)^2(y+1) + 2(y-1)^2 + 2 > 0$ for $y \geq -1$. Put $y = v/(2z)$ in (44) and extend the integral to ∞ . The added piece is at most 0 with q and at least 0 with p . Since $\frac{1}{\sqrt{\pi}} \int_0^\infty e^{-v} v^{k-1/2} dv = \Gamma(k + \frac{1}{2})/\Gamma(\frac{1}{2})$ equals $1, \frac{1}{2}, \frac{3}{4}, \frac{15}{8}$ for $k = 0, 1, 2, 3$, the 2 bounds integrate to the 2 sides of the lemma. The bound $E \leq 1$ follows in the same way from $\sqrt{1-y} \leq 1$. Finally $-\frac{3}{128z^2} + \frac{15}{512z^3} \leq 0$ exactly when $z \geq \frac{5}{4}$.

A.2 Proof of Theorem 3.4 for $N \geq 175$

Step 1: bounds that hold for every $\zeta \geq 4$. Fix $N \geq 175$ and write $\zeta = \zeta_N$, $U = \zeta + \frac{1}{8\zeta}$, $r = U/N$ and $n(\zeta) = \sqrt{8\zeta/\pi} e^\zeta$. Then $n(\zeta_N) = N$ by (8), and $\zeta \geq 4$ since $n(4) < 175$. For $k = 1, 2, 3$ the function $U^k/n(\zeta)$ decreases on $[4, \infty)$, because its logarithmic derivative is $k(1 - \frac{1}{8\zeta^2})/U - 1 - \frac{1}{2\zeta} < \frac{k}{U} - 1 < 0$. The same holds for U^4/n , since $U > 4$, and for ζ/n and ζ^2/n . The values at $\zeta = 4$ give, for every $\zeta \geq 4$,

$$r \leq 0.023135, \quad \frac{U^2}{N} \leq 0.093262, \quad \frac{U^3}{N} \leq 0.375960, \quad \frac{U^4}{N} \leq 1.51559, \quad \frac{\zeta(\zeta+1)}{N} \leq 0.114776.$$

Step 2: the upper bound. Let $W = U/(1-r)$, so that $NW/(N+W) = U$, and let $\delta = W(W+2)/(2N)$. By Lemma 3.5(b) and (9) it suffices to show that $G_U := \psi(W) - \psi(\zeta) + \log E(W) + \log(1-\delta) > 0$. We bound the 3 parts from below. First, $\psi(W) - \psi(\zeta) = (W-U) + (U-\zeta) + \frac{1}{2} \log \frac{U}{\zeta} + \frac{1}{2} \log \frac{W}{U}$, where $W-U = \frac{Ur}{1-r}$, $U-\zeta = \frac{1}{8\zeta}$, $\log \frac{U}{\zeta} = \log(1 + \frac{1}{8\zeta^2}) \geq \frac{1}{8\zeta^2} - \frac{1}{128\zeta^4}$ and $\log \frac{W}{U} = -\log(1-r) \geq r$. Second, Lemma 3.6 gives $E(W) \geq 1-t$ with $t = \frac{1}{8W} + \frac{3}{128W^2} + \frac{45}{512W^3}$. Since $W \geq U \geq \zeta$ and $\frac{1}{8U} = \frac{1}{8\zeta} - \frac{1}{8\zeta(8\zeta^2+1)} \leq \frac{1}{8\zeta} - \frac{1}{64\zeta^3} + \frac{1}{512\zeta^5}$, we get $t \leq \tau := \frac{1}{8\zeta} + \frac{3}{128\zeta^2} + \frac{37}{512\zeta^3} + \frac{1}{512\zeta^5}$ and $\log E(W) \geq \log(1-\tau) \geq -\tau - \frac{\tau^2}{2(1-\tau)}$. Third, $\delta = \frac{Ur}{2(1-r)^2} + \frac{r}{1-r}$ and $\log(1-\delta) \geq -\delta - \frac{\delta^2}{2(1-\delta)}$. Adding the 3 parts gives $G_U \geq B + D$, where B collects the terms in ζ alone and D the terms with a factor r :

$$B = \frac{5}{128\zeta^2} - \frac{37}{512\zeta^3} - \frac{1}{256\zeta^4} - \frac{1}{512\zeta^5} - \frac{\tau^2}{2(1-\tau)}, \quad D = \frac{r}{2(1-r)} \left(\frac{U(1-2r)}{1-r} - 1 - r \right) - \frac{\delta^2}{2(1-\delta)}.$$

Both $\zeta\tau$ and τ decrease in ζ , so $\zeta^2 B$ increases. At $\zeta = 4$ we have $\zeta\tau = 0.135384$, $\tau = 0.033846$ and $\zeta^2 B = 0.011236$. Hence $B > 0$. In D the bracket is at least $4.03125(1-0.023683) - 1.023135 > 2.912$, so the first term of D is at least $1.456r$. Write $\delta = rm$ with $m = \frac{U}{2(1-r)^2} + \frac{1}{1-r}$. Then $\delta \leq 0.07255$ and $rm^2 = \frac{U^3}{4N(1-r)^4} + \frac{U^2}{N(1-r)^3} + \frac{r}{(1-r)^2} \leq 0.2276$, so $\frac{\delta^2}{2(1-\delta)} = r \cdot \frac{rm^2}{2(1-\delta)} \leq 0.1228r$. Hence $D \geq 1.33r > 0$. So $G_U > 0$ and $z_N < U$.

Step 3: the lower bound. Let $\varepsilon = \varepsilon_N$, $\ell = U - \varepsilon$, $r' = \ell/N$ and $W' = \ell/(1-r')$, so that $NW'/(N+W') = \ell$. By Step 1, $\varepsilon \leq 0.114776 + \frac{1}{256} < 0.1187$, so $\ell \geq \zeta - 0.1187 \geq 3.88$ and $W' \geq \ell$. By Lemma 3.5(a) and (9) it suffices to show that $G_\ell := \psi(W') - \psi(\zeta) + \log E(W') < 0$. We bound the 2 parts from above. First, $\psi(W') - \psi(\zeta) = (\ell - \zeta) + \frac{\ell r'}{1-r'} + \frac{1}{2} \log \frac{\ell}{\zeta} - \frac{1}{2} \log(1-r')$, where $\ell - \zeta = \frac{1}{8\zeta} - \varepsilon$, $\frac{1}{2} \log \frac{\ell}{\zeta} \leq \frac{1}{2} \left(\frac{\ell}{\zeta} - 1 \right) = \frac{1}{16\zeta^2} - \frac{\varepsilon}{2\zeta}$ and $-\frac{1}{2} \log(1-r') \leq \frac{r'}{2(1-r')}$. The 2 terms in r' add up to $\frac{\ell^2 + \ell/2}{N(1-r')}$. Second, Lemma 3.6 gives $E(W') \leq 1-t'$ with $t' = \frac{1}{8W'} + \frac{3}{128W'^2} - \frac{15}{512W'^3} \in (0, 1)$,

so $\log E(W') \leq -t'$. Here $\frac{1}{8W'} = \frac{1}{8\ell} - \frac{1}{8N}$. Write $\ell = \zeta(1 + \kappa)$ with $\kappa = \frac{1}{8\zeta^2} - \frac{\varepsilon}{\zeta}$. Then $\frac{1}{\ell} \geq \frac{1-\kappa}{\zeta}$, so $-\frac{1}{8W'} \leq -\frac{1}{8\zeta} + \frac{1}{64\zeta^3} - \frac{\varepsilon}{8\zeta^2} + \frac{1}{8N}$. Next, $x \mapsto x/(1 - x/N)$ increases and $\ell \leq U$, so $W' \leq U/(1 - r)$ and $\frac{1}{W'^2} \geq \frac{(1-r)^2}{\zeta^2(1+\frac{1}{8\zeta^2})^2} \geq \frac{(1-2r)(1-\frac{1}{4\zeta^2})}{\zeta^2}$. Finally $\frac{15}{512W'^3} \leq \frac{15}{512\ell^3}$. Adding, and using $\varepsilon(1 + \frac{1}{2\zeta}) = \frac{\zeta^2 + \frac{3}{2}\zeta + \frac{1}{2}}{N} + \frac{1}{16\zeta^2} + \frac{1}{32\zeta^3}$, we get $G_\ell \leq \beta + \Delta - \frac{\varepsilon}{8\zeta^2}$ with

$$\beta = -\frac{3}{128\zeta^2}(1-2r)\left(1 - \frac{1}{4\zeta^2}\right) + \frac{15}{512\ell^3} - \frac{1}{64\zeta^3}, \quad \Delta = \frac{1}{N}\left(\frac{\ell^2 + \ell/2}{1-r'} + \frac{1}{8} - \zeta^2 - \frac{3}{2}\zeta - \frac{1}{2}\right).$$

For $\beta < 0$ it suffices that $\zeta^2/\ell^3 \leq \frac{4}{5}(1-2r)(1 - \frac{1}{4\zeta^2})$. The left side is at most $\zeta^2/(\zeta - 0.12)^3 \leq 16/3.88^3 < 0.274$, since this function decreases for $\zeta > 0.12$ (its logarithmic derivative is $\frac{2}{\zeta} - \frac{3}{\zeta-0.12} < 0$ there). The right side is at least $0.8 \cdot 0.9537 \cdot 0.9843 > 0.75$. For $\Delta < 0$, note that $\ell \leq U$ and $r' \leq r$, so

$$\frac{\ell^2 + \ell/2}{1-r'} \leq U^2 + \frac{U}{2} + \frac{U^3/N + U^2/(2N)}{1-r} \leq \zeta^2 + \frac{\zeta}{2} + 0.2667 + 0.4326,$$

using $U^2 + \frac{U}{2} = \zeta^2 + \frac{\zeta}{2} + \frac{1}{4} + \frac{1}{16\zeta} + \frac{1}{64\zeta^2}$. This is less than $\zeta^2 + \frac{3}{2}\zeta + \frac{3}{8}$ because $\zeta \geq 4$. Hence $G_\ell < 0$ and $z_N > \ell$.

The constants were recomputed in 40-digit arithmetic, and as a check outside the proof we evaluated $\zeta^2 B$, D , β and Δ on $\zeta \in [4, 60]$ with step 0.001.

A.3 The certificates

Let $c, d > 0$ be integers, and weight each repeat by d and each switch by c . By Theorem 2.1 and Proposition C.1 with $(\alpha, \beta) = (d, c)$,

$$d^{N-1}S_N(c/d) = \sum_j W(N, m_N, j) c^j d^{N-1-j} = n_N(m_N).$$

So $S_N(c/d) < 1$ exactly when the integer $n_N(m_N)$ is less than d^{N-1} . This is the identity used in Step 4 of the proof of Theorem 3.4.

A.4 Proof of Proposition 3.8

Step 1: an exact formula for R_N . Write $w = w_N$, $z = z_N$, $\Lambda_N = 2\Phi(w)/N$ and $\delta = w(w+2)/(2N)$. At $w = w_N$ the display in the proof of Lemma 3.5 gives $1 \leq \Lambda_N \leq 1/(1-\delta)$ when $\delta < 1$, and (9) gives $\psi(w) + \log E(w) = L + c_0 + \log \Lambda_N$. Since $w - z = wz/N$, $w/z = 1 + w/N$ and $\frac{1}{8z} = \frac{1}{8w} + \frac{1}{8N}$, the remainder $R_N = \psi(z) - L - c_0 - \frac{1}{8z}$ is

$$R_N = \log \Lambda_N - \left(\log E(w) + \frac{1}{8w}\right) - \frac{1}{8N} - \frac{wz}{N} - \frac{1}{2} \log\left(1 + \frac{w}{N}\right). \quad (45)$$

Step 2: the lower bound for $N \geq 175$. By the proof of Theorem 3.4, $3.88 \leq z \leq w \leq U/(1-r)$. Note that $\log \Lambda_N \geq 0$ and $-\log E(w) \geq -\log(1 - \frac{1}{8w}) \geq \frac{1}{8w}$ by Lemma 3.6, since $w \geq \frac{5}{4}$. So $R_N \geq -\frac{wz + w/2 + 1/8}{N} = -\frac{z^2 + z/2}{N-z} - \frac{1}{8N}$. This is at least $-\frac{z^2 + z}{N}$ as soon as $N \geq \frac{2z^2(z+1)}{z-1/4}$. The right side increases in z and $z \leq U$, so it suffices that $U^2/N \leq \frac{U-1/4}{2(U+1)}$. The left side is at most 0.0933 and the right side at least 0.3757.

Step 3: the upper bound for $N \geq 175$. Let $t = \frac{1}{8w} + \frac{3}{128w^2} + \frac{45}{512w^3} \leq \frac{1.09504}{8w}$, which holds as $w \geq 3.88$. Lemma 3.6 and $-\log(1-t) \leq t + \frac{t^2}{2(1-t)}$ give

$$-\log E(w) - \frac{1}{8w} \leq \frac{3}{128w^2} + \frac{45}{512w^3} + \frac{t^2}{2(1-t)} \leq \frac{0.0559}{w^2} \leq \frac{0.0559}{z^2}.$$

Using $\log \Lambda_N \leq \delta + \frac{\delta^2}{2(1-\delta)}$, $\log(1+x) \geq x - \frac{x^2}{2}$ and $wz \geq w^2 - w^3/N$, the rest of (45) is at most $-\frac{(w-1/2)^2}{2N} + \frac{w^3+w^2/4}{N^2} + \frac{\delta^2}{2(1-\delta)}$. By the bounds on r and U^k/N in Step 1 of the previous proof, with $w \leq U/(1-r)$, we have $\frac{w^3+w^2/4}{N} \leq 0.4278$ and $N\frac{\delta^2}{2(1-\delta)} = \frac{(w^4+4w^3+4w^2)/(4N)}{2(1-\delta)} \leq 0.4945$, while $\frac{(w-1/2)^2}{2} \geq 5.71$. So the rest is negative and $R_N \leq 0.0559/z^2 < \frac{1}{16z^2}$.

Step 4: the form in L for $N \geq 175$. We have $z < U = L + c_0 - \frac{1}{2} \log \zeta + \frac{1}{8\zeta} < L$. Also $z > \zeta - 0.1187 \geq \nu_N - 0.1187 \geq L/2$, because $L - \log L \geq 1.18$ for $L \geq \log 175$. Then $\frac{1}{16z^2} \leq \frac{1}{4L^2}$ and $\frac{z^2+z}{N} \leq \frac{L^2+L}{N}$.

Step 5: $N \leq 174$. The function $R(z) = \psi(z) - L - c_0 - \frac{1}{8z}$ increases in z , while $\frac{1}{16z^2}$ and $-\frac{z^2+z}{N}$ decrease. So it suffices to check $R(z(c'/d)) \leq \frac{1}{16z(c'/d)^2}$ and $R(z(c/d)) \geq -\frac{z(c/d)^2+z(c/d)}{N}$ with the certificates, together with $z(c/d) \geq L/2$ and, for $N \geq 3$, $z(c'/d) \leq L$. All hold, and the smallest gaps are 0.0374 (upper) and 0.0813 (lower), both at $N = 174$. For $N = 2$ we have $S_2(u) = 2u$, so $u_2 = \frac{1}{2}$ and $z_2 = \frac{2}{3}$, and $\frac{2}{3} < \log 2$ is equivalent to $e^2 < 8$, which holds since $e^2 < 2.72^2 = 7.3984$.

B The uniform local law

We keep the notation of Section 6, with $\tilde{I}_j(x) = e^{-x} I_j(x)$. The first 2 lemmas set up the tilt, the third writes both $P_N(k)$ and $U_N(k)$ as Fourier integrals, the next 2 compare the modulus and the phase of the 2 integrands, and the last computes the moments of κ that appear in the bound.

Lemma B.1 (tilting). *For $s > 0$ let $Y_1 \sim \text{Bin}(n_1, r_1)$ and $Y_2 \sim \text{Bin}(n_2, r_2)$ be independent under $\mathbb{P}_s$. Then $\mathbb{P}(D = j) = \Lambda(s)s^{-j}\mathbb{P}_s(D = j)$ for every $j \in \mathbb{Z}$.*

Proof. Note that $\mathbb{P}(Y_1 = r)s^r = \binom{n_1}{r}(qs)^r p^{n_1-r} = (p + qs)^{n_1} \mathbb{P}_s(Y_1 = r)$, and likewise $\mathbb{P}(Y_2 = r)s^{-r} = (p + q/s)^{n_2} \mathbb{P}_s(Y_2 = r)$. Hence $\mathbb{P}(D = j) = s^{-j} \sum_r \mathbb{P}(Y_1 = r)s^r \mathbb{P}(Y_2 = r - j)s^{j-r} = \Lambda(s)s^{-j}\mathbb{P}_s(D = j)$. $\square$

Lemma B.2 (saddle). *If $n_1, n_2 \geq 1$, then $\mathbb{E}_s D = n_1 r_1 - n_2 r_2$ increases strictly from $-n_2$ to n_1 as s goes from 0 to ∞ . So it vanishes at a unique $s > 0$, the positive root of $n_1 p s^2 + (n_1 - n_2) q s - n_2 p = 0$, and $s \geq 1$ if and only if $n_1 \leq n_2$.*

Proof. Note that r_1 increases from 0 to 1 and r_2 decreases from 1 to 0. Clearing denominators in $n_1 q s / (p + q s) = n_2 q / (p s + q)$ gives the quadratic, and at $s = 1$ the difference is $(n_1 - n_2)q$. $\square$

From now on $2 \leq k \leq N - 2$ and s is this root.

Lemma B.3 (Fourier form). *Let $\chi(t) = \mathbb{E}_s e^{itD} = (1 - r_1 + r_1 e^{it})^{n_1} (1 - r_2 + r_2 e^{-it})^{n_2}$, $g(t) = e^{x(\cos t - 1)}$ and $\omega(t) = 2 + s e^{it} + s^{-1} e^{-it}$. Then*

$$P_N(k) = \frac{q}{2} \Lambda(s) \frac{1}{2\pi} \int_{-\pi}^{\pi} \chi \omega dt, \quad U_N(k) = \frac{q}{2} \Lambda(s) \frac{1}{2\pi} \int_{-\pi}^{\pi} g \omega dt.$$

Proof. By Theorem 6.1 and Lemma B.1, $P_N(k) = \frac{q}{2}\Lambda(s)[2\mathbb{P}_s(D=0) + s^{-1}\mathbb{P}_s(D=1) + s\mathbb{P}_s(D=-1)]$, and $\mathbb{P}_s(D=j) = \frac{1}{2\pi} \int \chi e^{-ijt} dt$. Since g is even and $I_j(x) = \frac{1}{\pi} \int_0^\pi e^{x \cos t} \cos(jt) dt$ [28, Eq. 10.32.3], we have $\frac{1}{2\pi} \int g dt = \tilde{I}_0(x)$ and $\frac{1}{2\pi} \int g e^{\pm it} dt = \tilde{I}_1(x)$. $\square$

Lemma B.4 (modulus). *We have $|\chi| \leq g$ and $g - |\chi| \leq g \min(1, 4\kappa^2 V)$.*

Proof. Note that $|1 - r + re^{\pm it}|^2 = 1 - 2r(1 - r)\kappa$. Put $f(z) = \sqrt{1 - z} e^{z/2}$ and $z_i = 2v_i \kappa \in [0, 1]$ ($v_i \leq \frac{1}{4}$, $\kappa \leq 2$). As $x = n_1 v_1 + n_2 v_2$, we get $|\chi|/g = f(z_1)^{n_1} f(z_2)^{n_2}$. Now $f(0) = 1$ and $f'(z) = -\frac{z}{2} e^{z/2} (1 - z)^{-1/2} \leq 0$, so $0 \leq f \leq 1$. Also $f(z) \geq 1 - z^2$, since for $z < 1$ this says $e^z \geq (1 - z)(1 + z)^2 = 1 + z - z^2 - z^3$, which follows from $e^z \geq 1 + z$. So Lemma 3.2(a), with the $n_1 + n_2$ numbers $1 - f(z_i) \leq z_i^2$, gives $1 - |\chi|/g \leq n_1 z_1^2 + n_2 z_2^2 = 4\kappa^2 V$. $\square$

Lemma B.5 (phase). *For $|t| < \pi$ write $\chi(t) = |\chi(t)|e^{i\Theta(t)}$ with Θ continuous and $\Theta(0) = 0$. Then $|\Theta(t)| \leq c_\varphi M|t|^3$ with $c_\varphi = 128/(3\pi^3) = 1.3761\dots$*

Proof. For $0 \leq r \leq 1$ and $|t| < \pi$, $w_r(t) = 1 - r + re^{it}$ lies on the segment from 1 to e^{it} , so $w_r \neq 0$ and $\arg w_r(t)$ lies between 0 and t . Writing $w_r(t) = e^{it/2} [\cos \frac{t}{2} + i(2r - 1) \sin \frac{t}{2}]$ gives $\arg w_r(t) = \frac{t}{2} - \arctan((1 - 2r) \tan \frac{t}{2})$. As the factor for $-Y_2$ is $\overline{w_{r_2}}$, $\Theta = n_1 \arg w_{r_1} - n_2 \arg w_{r_2}$. Subtracting $(n_1 r_1 - n_2 r_2) \sin t = 0$ gives $\Theta = n_1 \varphi(r_1, t) - n_2 \varphi(r_2, t)$ with $\varphi(r, t) = \arg w_r(t) - r \sin t$. Both are odd in t , so let $0 < t < \pi$. We show that $0 \leq \varphi(r, t) \leq c_\varphi r^2 t^3$.

Put $\tau = \tan \frac{t}{2}$, $b_0 = 1 - 2r$ and $\gamma(b) = \arctan(b\tau) - b\tau/(1 + \tau^2)$. Since $\frac{t}{2} = \arctan \tau$ and $\sin t = 2\tau/(1 + \tau^2)$, we get $\varphi = \gamma(1) - \gamma(b_0)$. Also γ is odd and $\gamma'(b) = \tau^3(1 - b^2)/[(1 + \tau^2)(1 + b^2\tau^2)] \geq 0$ on $[-1, 1]$. Hence

$$\varphi = \frac{\tau^3}{1 + \tau^2} \int_{b_0}^1 \frac{1 - b^2}{1 + b^2\tau^2} db \geq 0, \quad \int_{b_0}^1 (1 - b^2) db = 4r^2 - \frac{8}{3}r^3 \leq 4r^2.$$

There are 4 cases. (a) If $t \leq \frac{\pi}{2}$, then $\tau \leq 2t/\pi \leq 1$ by convexity of $\tan$ on $[0, \frac{\pi}{4}]$, so $\varphi \leq 4r^2\tau^3 \leq \frac{32}{\pi^3}r^2t^3$. (b) If $t > \frac{\pi}{2}$ and $r \leq \frac{1}{4}$, then $b_0 \geq \frac{1}{2}$, so $1 + b^2\tau^2 \geq 1 + \tau^2/4$ and $\varphi \leq 16r^2\tau^3/[(1 + \tau^2)(4 + \tau^2)]$. As $(1 + \tau^2)(4 + \tau^2) \geq \max(9\tau^2, \tau^4)$, this is at most $16r^2 \min(\tau/9, 1/\tau) \leq \frac{16}{3}r^2 \leq c_\varphi r^2 t^3$, as $t^3 > \pi^3/8$. (c) If $t > \frac{\pi}{2}$ and $\frac{1}{4} < r \leq \frac{1}{2}$, then $\gamma(b_0) \geq \gamma(0) = 0$, so $\varphi \leq \gamma(1) = \frac{1}{2}(t - \sin t) \leq \frac{t^3}{12} < \frac{4}{3}r^2 t^3$. (d) If $r > \frac{1}{2}$, then $\varphi = \gamma(1) + \gamma(|b_0|) \leq 2\gamma(1) = t - \sin t \leq \frac{t^3}{6} < \frac{2}{3}r^2 t^3$. As $\frac{32}{\pi^3}, \frac{4}{3}, \frac{2}{3} < c_\varphi$, the claim follows. Finally $\varphi(r_1, t)$ and $\varphi(r_2, t)$ have the same sign, so $|\Theta| \leq \max_i n_i \varphi(r_i, t) \leq c_\varphi |t|^3 \max_i n_i r_i^2 = c_\varphi M|t|^3$. $\square$

Lemma B.6 (moments). *For every $x > 0$, with $\mathcal{R} = \mathcal{R}(x)$,*

- (i) $\langle \kappa^2 \rangle = 2 - 2\mathcal{R} - \mathcal{R}/x$ and $\langle \kappa^3 \rangle = 4 - 4\mathcal{R} - 3\mathcal{R}/x - 2\mathcal{R}/x^2 + 1/x$,
- (ii) $1/\mathcal{R}(x) \leq 1 + 2/x$,
- (iii) $\langle \kappa^2 \rangle \leq \min(\frac{3}{2}, 2/x^2)$ and $\langle \kappa^3 \rangle \leq \min(\frac{5}{2}, 8/x^3)$.

Proof. (i) Note that $\kappa^2 = \frac{3}{2} - 2\cos t + \frac{1}{2}\cos 2t$, $\kappa^3 = \frac{5}{2} - \frac{15}{4}\cos t + \frac{3}{2}\cos 2t - \frac{1}{4}\cos 3t$ and $\langle \cos jt \rangle = I_j(x)/I_0(x)$ [28, Eq. 10.32.3]. The recurrence $I_{\nu-1} - I_{\nu+1} = (2\nu/x)I_\nu$ [28, Eq. 10.29.1] gives $I_2/I_0 = 1 - 2\mathcal{R}/x$ and $I_3/I_0 = \mathcal{R} - 4/x + 8\mathcal{R}/x^2$.

(ii), (iii) By [28, Eqs. 10.29.2, 10.29.3], $I'_0 = I_1$ and $I'_1 = I_0 - I_1/x$, so $\mathcal{R}' = 1 - \mathcal{R}/x - \mathcal{R}^2$. For a C^1 function h put $d_h = 1 - h/x - h^2 - h'$. If $d_h > 0$ on $[x_*, \infty)$ and $\mathcal{R}(x_*) > h(x_*)$, then $\mathcal{R} > h$ on $[x_*, \infty)$. Indeed, at a first x_1 with $\mathcal{R}(x_1) = h(x_1)$ we would have $(\mathcal{R} - h)'(x_1) \leq 0$, but it is $d_h(x_1) > 0$. Each d_h below follows by clearing denominators.

First, $h_0 = x/(x+2)$ has $d_{h_0} = 3x/(x+2)^2 > 0$, and $\mathcal{R} - h_0 = \frac{x^2}{4} + O(x^3)$ as $x \rightarrow 0$ since $\mathcal{R} = \frac{x}{2} - \frac{x^3}{16} + O(x^5)$. So $\mathcal{R} > h_0$ for all $x > 0$, which is (ii). It also gives $\langle \kappa^2 \rangle \leq 2 - (2 + \frac{1}{x})h_0 = \frac{3}{x+2} \leq \frac{3}{2}$.

Second, by (i), $\langle \kappa^2 \rangle < 2/x^2$ means $\mathcal{R} > h_2 = (2x^2 - 2)/(x(2x + 1))$. Here $d_{h_2} = (5x^2 - 4x - 4)/(x^2(2x + 1)^2) > 0$ for $x > 1.3798$, and at $x_* = 1.38$ we have $\mathcal{R} > h_0 = 0.40828 > h_2 = 0.34860$. For $x < 1.38$, $\frac{3}{x+2} \leq \frac{2}{x^2}$ since $3x^2 - 2x - 4 \leq 0$ for $x \leq 1.535$.

Third, $\langle \kappa^3 \rangle < 8/x^3$ means $\mathcal{R} > h_3 = (4x^3 + x^2 - 8)/(x(4x^2 + 3x + 2))$. Here $d_{h_3} = (49x^3 - 48x^2 - 24x - 64)/(x^2(4x^2 + 3x + 2)^2) > 0$ for $x \geq 1.72$, and at $x_* = 1.72$ we have $\mathcal{R} > h_2 = 0.51289 > h_3 = 0.46870$. For $x < 1.72$, $\langle \kappa^3 \rangle \leq 2\langle \kappa^2 \rangle \leq 4/x^2 \leq 8/x^3$.

Finally, the density is proportional to $e^{-x\kappa}$, so $\frac{d}{dx}\langle \kappa^j \rangle = \langle \kappa^j \rangle \langle \kappa \rangle - \langle \kappa^{j+1} \rangle \leq 0$ by Chebyshev's sum inequality for the increasing functions κ^j and κ . As $x \rightarrow 0$ the averages tend to $\frac{1}{2\pi} \int (1 - \cos t)^2 dt = \frac{3}{2}$ and $\frac{1}{2\pi} \int (1 - \cos t)^3 dt = \frac{5}{2}$. $\square$

Proof of Theorem 6.2. Step 1: the error as an integral. By Lemma B.3, $\theta = \Delta/(\tilde{I}_0 + (s + s^{-1})\tilde{I}_1)$ with $\Delta = \frac{1}{2\pi} \int (\chi - g)\omega dt$. Both integrals there are real, so Δ integrates

$$\operatorname{Re}[(\chi - g)\omega] = (|\chi| \cos \Theta - g)(2 + (s + s^{-1}) \cos t) - |\chi| \sin \Theta (s - s^{-1}) \sin t.$$

Step 2: bounding the integrand. Now $g - |\chi| \cos \Theta = (g - |\chi|) + |\chi|(1 - \cos \Theta)$ lies between 0 and $g\mu$ with $\mu = \min(1, 4\kappa^2 V) + \frac{1}{2}\Theta^2$, by Lemma B.4 and $1 - \cos \Theta \leq \frac{1}{2}\Theta^2$. Also $|\chi \sin \Theta| \leq g|\Theta|$ and $|s - s^{-1}| \leq s + s^{-1}$. Since $\frac{1}{2\pi} \int gh dt = \tilde{I}_0 \langle h \rangle$,

$$|\Delta| \leq 2\tilde{I}_0 \langle \mu \rangle + (s + s^{-1})\tilde{I}_0 [\langle \mu | \cos t \rangle + \langle |\Theta| |\sin t| \rangle].$$

Since $|\cos t| \leq 1$ and $\tilde{I}_0/\tilde{I}_1 = 1/\mathcal{R} \geq 1$, the mediant inequality gives

$$|\theta| \leq \mathcal{E}_0 := \frac{\langle \min(1, 4\kappa^2 V) \rangle + \frac{1}{2}\langle \Theta^2 \rangle + \langle |\Theta| |\sin t| \rangle}{\mathcal{R}(x)}.$$

Step 3: from $\mathcal{E}_0$ to $\mathcal{E}$. For $\mathcal{E}_0 \leq \mathcal{E}$ we use $\min(1, 4\kappa^2 V) \leq 4\kappa^2 V$, Lemma B.5, $|t| \leq \pi |\sin \frac{t}{2}| = \pi(\kappa/2)^{1/2}$ and $|\sin t| \leq 2(\kappa/2)^{1/2}$. Since $c_\varphi \pi^3 = \frac{128}{3}$, we get $\frac{1}{2}\Theta^2 \leq \frac{1}{2}c_\varphi^2 \pi^6 M^2 (\kappa/2)^3 = \frac{1024}{9} M^2 \kappa^3$ and $|\Theta| |\sin t| \leq 2c_\varphi \pi^3 M (\kappa/2)^2 = \frac{64}{3} M \kappa^2$. $\square$

Proof of Corollary 6.3. Step 1: bounds on V and M . Note that $r_1 = r_2 = q$ at $s = 1$, $r_2 \leq q$ if $s \geq 1$, and $r_1 \leq q$ if $s \leq 1$. So $\min(r_1, r_2) \leq q \leq \frac{1}{2}$ and $x = m(1 - r_1) + m(1 - r_2) \geq m/2$. Also $n_i v_i \leq x$, so $v_i \leq \min(x/n_{\min}, r_{\max})$ and $V \leq x \max_i v_i \leq \min(x^2/n_{\min}, x r_{\max})$. Since $r_i = m/n_i$, $M = m r_{\max} = m^2/n_{\min} \leq \min(4x^2/n_{\min}, 2x r_{\max})$. We put these and Lemma B.6 into $\mathcal{E}$.

Step 2: the $n_{\min}$ form. The first term of $\mathcal{E}$ is at most $\frac{4+256/3}{n_{\min}}(x^2 + 2x) \min(\frac{3}{2}, \frac{2}{x^2}) \leq \frac{(4+256/3)(2+2\sqrt{3})}{n_{\min}} \leq \frac{488.13}{n_{\min}}$, with the supremum at $x^2 = \frac{4}{3}$ (the product is 488.126...). If $x \leq 1$, the M^2 term is at most $\frac{3}{x} \cdot \frac{1024}{9} \cdot \frac{16x^4}{n_{\min}^2} \cdot \frac{5}{2} \leq \frac{13654}{n_{\min}^2}$. If $x \geq 1$, then $M^2 \leq \min(16x^4/n_{\min}^2, 4x^2)$, and the term is at most

$$3 \cdot \frac{1024}{9} \min\left(\frac{16x^4}{n_{\min}^2}, 4x^2\right) \frac{8}{x^3} = \frac{1024}{3} \min\left(\frac{128x}{n_{\min}^2}, \frac{32}{x}\right) \leq \frac{1024}{3} \cdot \frac{64}{n_{\min}} = \frac{65536/3}{n_{\min}} \leq \frac{21845.34}{n_{\min}},$$

since a minimum is at most the geometric mean. So both cases give at most $21845.34/n_{\min}$, and $488.13 + 21845.34 = 22333.47 \leq 22334$.

Step 3: the $r_{\max}$ form. The first term is at most $(4 + \frac{128}{3})r_{\max} \sup_x (x + 2) \min(\frac{3}{2}, \frac{2}{x^2}) = (4 + \frac{128}{3})(3 + \sqrt{3})r_{\max} \leq 220.83 r_{\max}$ (the product is 220.829..., with the supremum at $x^2 = \frac{4}{3}$). The M^2 term is at most $\frac{4096}{9} r_{\max}^2 \sup_x (x^2 + 2x) \min(\frac{5}{2}, \frac{8}{x^3})$, where the supremum is 12.7968... at $x^3 = 3.2$. So the term is at most $\frac{4096}{9} \cdot 12.797 r_{\max}^2 \leq 5825 r_{\max}^2$.

Step 4: $r_{\max} \leq \delta_T$. By symmetry let $n_1 \leq n_2$, and put $u = q/p \leq 2q$. Then $s \geq 1$, so $r_{\max} = r_1 \leq qs/p = us$. By Lemma B.2,

$$s = \frac{(n_2 - n_1)u}{2n_1} + \sqrt{\frac{(n_2 - n_1)^2 u^2}{4n_1^2} + \frac{n_2}{n_1}} \leq \frac{(n_2 - n_1)u}{n_1} + \sqrt{\frac{n_2}{n_1}}.$$

With $n_1 \geq 1$ and $n_2 \leq N$ this gives $r_{\max} \leq 2q\sqrt{N} + 4q^2N = \delta_T$. $\square$

C The numerator tables

This appendix answers Kagey's question about his tables. We first give the generating function of the numerators for any rational p , and then the 2 bijections that explain the tables for $p = \frac{2}{3}$ and $p = \frac{1}{3}$.

Proposition C.1. *For $p = \alpha/(\alpha + \beta)$ and $N \geq 1$, the numerators of Remark 2.3 are $n_N(k) = \sum_w \alpha^{\text{rep}(w)} \beta^{\text{chg}(w)}$, summed over the words with k letters R and $N - k$ letters L. So they are integers, and*

$$\sum_{a,b \geq 0} n_{a+b}(a) x^a y^b = \frac{1 - (\alpha - 1)(x + y) + (\alpha - \beta)(\alpha + \beta - 2)xy}{1 - \alpha(x + y) + (\alpha^2 - \beta^2)xy}.$$

Proof. Multiply (1) by $2(\alpha + \beta)^{N-1}$. Let H_R and H_L be the weighted sums over the words ending in R and in L, with x marking R and y marking L. Then $H_R = x(1 + \alpha H_R + \beta H_L)$ and $H_L = y(1 + \alpha H_L + \beta H_R)$, and solving as in the proof of Proposition 2.2 gives $1 + H_R + H_L$. $\square$

For $(\alpha, \beta) = (2, 1)$ this is $(1 - x)(1 - y)/(1 - 2x - 2y + 3xy)$, and for $(1, 2)$ it is $(1 - xy)/(1 - x - y - 3xy)$, the generating functions recorded in A035002 and A348595 [22, 23]. Figure 6 shows both bijections.

Theorem C.2 ($p = \frac{2}{3}$ and rook paths). *Let $r(a, b)$ count the rook paths from $(0, 0)$ to (a, b) with every move to the right or up. For $p = \frac{2}{3}$, $n_N(k) = r(k, N - k)$ for all $0 \leq k \leq N$. Since the array of A035002 starts at $(1, 1)$, Kagey's $p = \frac{2}{3}$ table read by rows is A035002 read by antidiagonals.*

Proof. By definition each entry of A035002 is the sum of the entries to its left and above it, which is the recursion for r obtained by removing the last move. A rook path to (a, b) with $a + b = N \geq 1$ gives a word w with a letters R and b letters L, one letter per unit of length, and a set of cuts $i < N$ where one move ends and the next begins. A cut is forced where $w_i \neq w_{i+1}$ and optional where $w_i = w_{i+1}$, and each choice of optional cuts gives exactly one rook path. Hence, $r(a, b) = \sum_w 2^{\text{rep}(w)}$, which is $n_N(a)$ for $(\alpha, \beta) = (2, 1)$ by Proposition C.1. $\square$

Theorem C.3 ($p = \frac{1}{3}$ and Mathar's blocked walks). *Call (x, y) blocked if $x \equiv y \equiv 1$ or $x \equiv y \equiv 2 \pmod{3}$, and let $M(a, b)$ count the walks from $(0, 0)$ to $(3a, 3b)$ with unit steps right and up that avoid blocked points. For $p = \frac{1}{3}$, $n_N(k) = M(k, N - k)$. The array M is symmetric.*

Proof. Work modulo 3 and call a point with residue $(0, 0)$ a *corner*. Since $(1, 1)$ and $(2, 2)$ are blocked, a right step from a corner is followed by another, to $(2, 0)$, which we call state A, and an up step leads on to $(0, 2)$, state B. From A the walk steps right to a corner, or it goes up, right and up to B, and B behaves symmetrically. Leaving A crosses one line $x \equiv 0$ and no line $y \equiv 0$, leaving B does the reverse, and leaving a corner crosses neither. A walk to $(3a, 3b)$ crosses a lines $x \equiv 0$

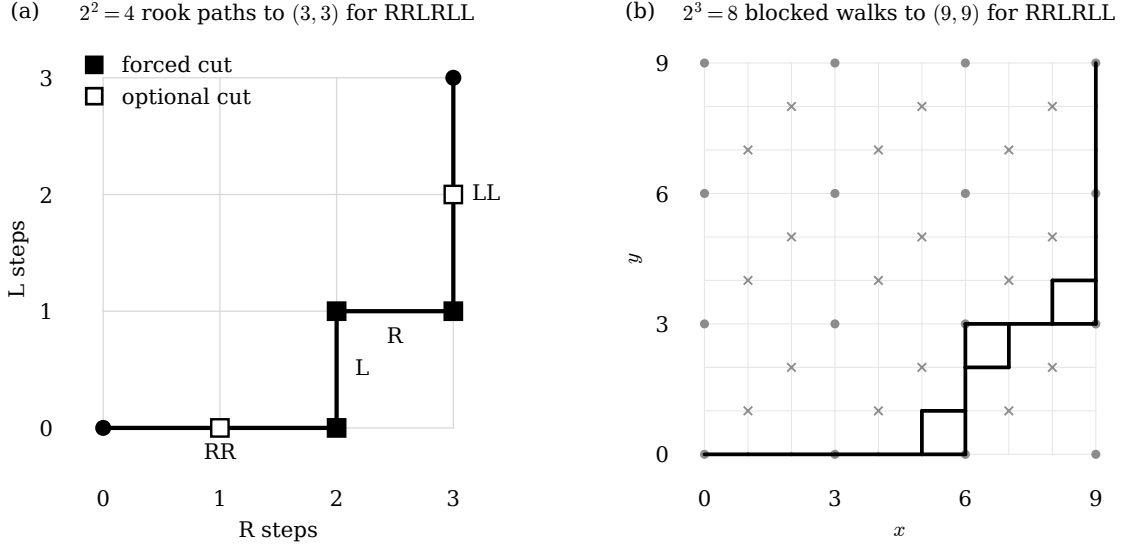


Figure 6: The 2 bijections for the word RRLRLL. (a) The rook path. A cut is forced at each of the 3 direction changes (filled) and optional between equal letters (hollow), so the word comes from $2^2 = 4$ rook paths to $(3, 3)$. (b) The union of the $2^3 = 8$ blocked walks to $(9, 9)$ that give the word, with 1 detour square for each direction change. Blocked points are marked $\times$ and corners by dots.

and b lines $y \equiv 0$, so writing R for each visit to A and L for each visit to B gives a word w with a letters R and b letters L. Between w_i and w_{i+1} the walk passes through a corner if $w_i = w_{i+1}$, in 1 way, and through a corner or directly if $w_i \neq w_{i+1}$, in 2 ways. The start and the end are forced. Hence $M(a, b) = \sum_w 2^{\text{chg}(w)}$, which is $n_N(a)$ for $(\alpha, \beta) = (1, 2)$ by Proposition C.1. Swapping the letters R and L keeps $\text{chg}(w)$, so $M(a, b) = M(b, a)$. $\square$

In Mathar’s note [21] the “squares” of A348595 are lattice points. A direct count of the walks gives its published terms (computer-checked, Section 7). So A348595 lists the lower triangle of M , which is how that entry is “related” to the $p = \frac{1}{3}$ table.

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